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CHAPTER

11 BJT and JFET Frequency Response

11.1 INTRODUCTION

The analysis thus far has been limited to a particular frequency. For the amplifier, it was a frequency that normally permitted ignoring the effects of the capacitive elements, reducing the analysis to one that included only resistive elements and sources of the independent and controlled variety. We will now investigate the frequency effects introduced by the larger capacitive elements of the network at low frequencies and the smaller capacitive elements of the active device at the high frequencies. Since the analysis will extend through a wide frequency range, the logarithmic scale will be defined and used throughout the analysis. In addition, since industry typically uses a decibel scale on its frequency plots, the concept of the decibel is introduced in some detail. The similarities between the frequency response analyses of both BJTs and FETs permit a coverage of each in the same chapter.

11.2 LOGARITHMS

There is no escaping the need to become comfortable with the logarithmic function. The plotting of a variable between wide limits, comparing levels without unwieldy numbers, and identifying levels of particular importance in the design, review, and analysis procedures are all positive features of using the logarithmic function.

As a first step in clarifying the relationship between the variables of a logarithmic function, consider the following mathematical equations:

$$a = b^x, \qquad x = \log_b a \tag{11.1}$$

The variables a, b, and x are the same in each equation. If a is determined by taking the base b to the x power, the same x will result if the log of a is taken to the base b. For instance, if $b = 10$ and $x = 2$,

$$a = b^x = (10)^2 = 100$$

but

$$x = \log_b a = \log_{10} 100 = 2$$

In other words, if you were asked to find the power of a number that would result in a particular level such as shown below:

$$10{,}000 = 10^x$$

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the level of x could be determined using logarithms. That is,

$$x = \log_{10} 10{,}000 = 4$$

For the electrical/electronics industry and in fact for the vast majority of scientific research, the base in the logarithmic equation is limited to 10 and the number $e = 2.71828. . . .$

Logarithms taken to the base 10 are referred to as *common logarithms*, while logarithms taken to the base e are referred to as *natural logarithms*. In summary:

$$\text{Common logarithm:} \quad x = \log_{10} a \tag{11.2}$$

$$\text{Natural logarithm:} \quad y = \log_e a \tag{11.3}$$

The two are related by

$$\log_e a = 2.3 \log_{10} a \tag{11.4}$$

On today's scientific calculators, the common logarithm is typically denoted by the log key and the natural logarithm by the ln key.

EXAMPLE 11.1

Using the calculator, determine the logarithm of the following numbers to the base indicated.
(a) $\log_{10} 10^6$.
(b) $\log_e e^3$.
(c) $\log_{10} 10^{-2}$.
(d) $\log_e e^{-1}$.

Solution

(a) **6** (b) **3** (c) **−2** (d) **−1**

The results in Example 11.1 clearly reveal that the logarithm of a number taken to a power is simply the power of the number if the number matches the base of the logarithm. In the next example, the base and the variable x are not related by an integer power of the base.

EXAMPLE 11.2

Using the calculator, determine the logarithm of the following numbers.
(a) $\log_{10} 64$.
(b) $\log_e 64$.
(c) $\log_{10} 1600$.
(d) $\log_{10} 8000$.

Solution

(a) **1.806** (b) **4.159** (c) **3.204** (d) **3.903**

Note in parts (a) and (b) of Example 11.2 that the logarithms $\log_{10} a$ and $\log_e a$ are indeed related as defined by Eq. (11.4). In addition, note that the logarithm of a number does not increase in the same linear fashion as the number. That is, 8000 is 125 times larger than 64, but the logarithm of 8000 is only about 2.16 times larger

than the magnitude of the logarithm of 64, revealing a very nonlinear relationship. In fact, Table 11.1 clearly shows how the logarithm of a number increases only as the exponent of the number. If the antilogarithm of a number is desired, the 10^x or e^x calculator functions are employed.

TABLE 11.1

$\log_{10} 10^0$	$= 0$
$\log_{10} 10$	$= 1$
$\log_{10} 100$	$= 2$
$\log_{10} 1{,}000$	$= 3$
$\log_{10} 10{,}000$	$= 4$
$\log_{10} 100{,}000$	$= 5$
$\log_{10} 1{,}000{,}000$	$= 6$
$\log_{10} 10{,}000{,}000$	$= 7$
$\log_{10} 100{,}000{,}000$	$= 8$
and so on	

EXAMPLE 11.3

Using a calculator, determine the antilogarithm of the following expressions:
(a) $1.6 = \log_{10} a$.
(b) $0.04 = \log_e a$.

Solution

(a) $a = 10^{1.6}$
Calculator keys: 1 . 6 2nd F 10^x
and $a =$ **39.81**

(b) $a = e^{0.04}$
Calculator keys: 0 . 0 4 2nd F e^x
and $a =$ **1.0408**

Since the remaining analysis of this chapter employs the common logarithm, let us now review a few properties of logarithms using solely the common logarithm. In general, however, the same relationships hold true for logarithms to any base.

$$\log_{10} 1 = 0 \tag{11.5}$$

As clearly revealed by Table 11.1, since $10^0 = 1$,

$$\log_{10} \frac{a}{b} = \log_{10} a - \log_{10} b \tag{11.6}$$

which for the special case of $a = 1$ becomes

$$\log_{10} \frac{1}{b} = -\log_{10} b \tag{11.7}$$

revealing that for any b greater than 1 the logarithm of a number less than 1 is always negative.

$$\log_{10} ab = \log_{10} a + \log_{10} b \tag{11.8}$$

In each case, the equations employing natural logarithms will have the same format.

EXAMPLE 11.4

Using a calculator, determine the logarithm of the following numbers:

(a) $\log_{10} 0.5$.

(b) $\log_{10} \dfrac{4000}{250}$.

(c) $\log_{10} (0.6 \times 30)$.

Solution

(a) $\mathbf{-0.3}$

(b) $\log_{10} 4000 - \log_{10} 250 = 3.602 - 2.398 = \mathbf{1.204}$

Check: $\log_{10} \dfrac{4000}{250} = \log_{10} 16 = \mathbf{1.204}$

(c) $\log_{10} 0.6 + \log_{10} 30 = -0.2218 + 1.477 = \mathbf{1.255}$

Check: $\log_{10} (0.6 \times 30) = \log_{10} 18 = \mathbf{1.255}$

The use of log scales can significantly expand the range of variation of a particular variable on a graph. Most graph paper available is of the semilog or double-log (log-log) variety. The term *semi* (meaning one-half) indicates that only one of the two scales is a log scale, whereas double-log indicates that both scales are log scales. A semilog scale appears in Fig. 11.1. Note that the vertical scale is a linear scale with equal divisions. The spacing between the lines of the log plot is shown on the graph.

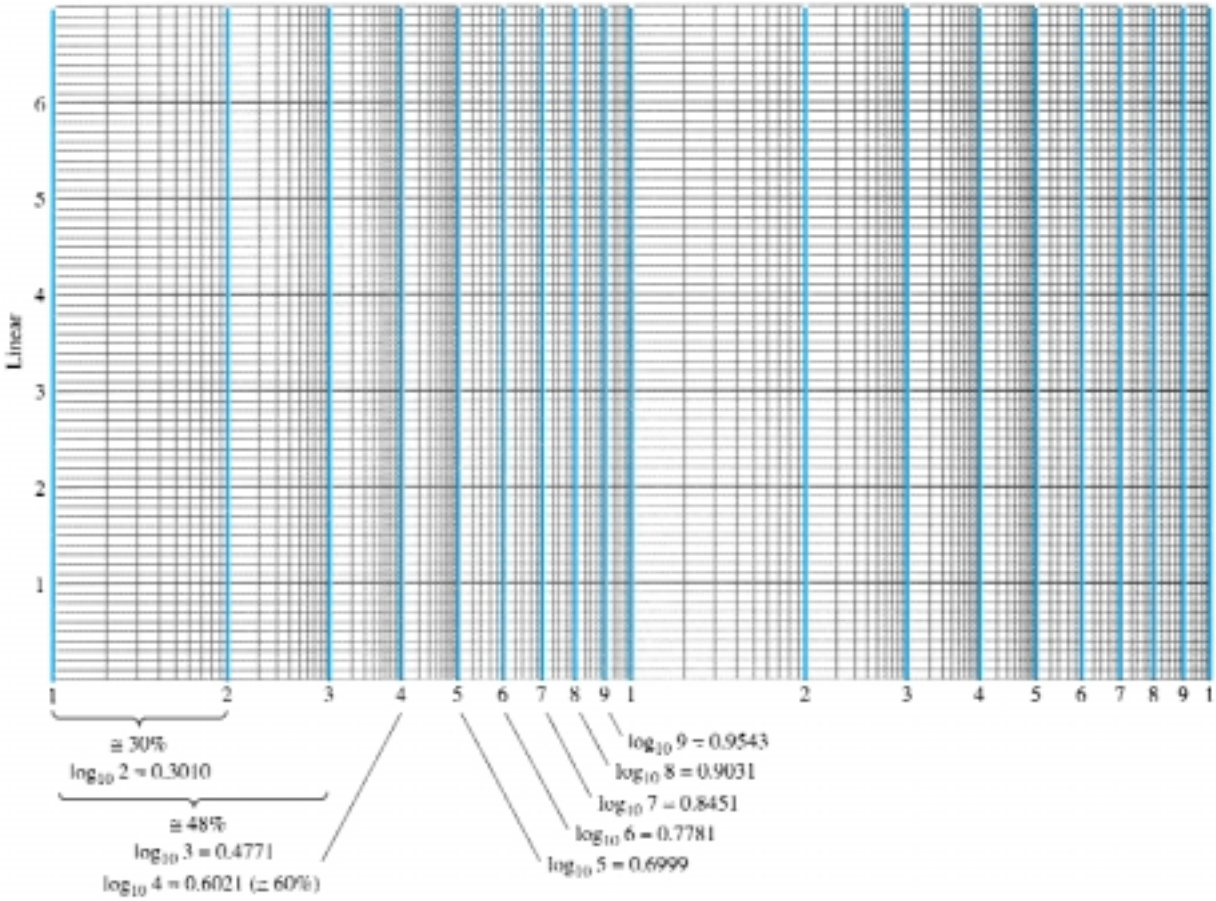


Figure 11.1 Semilog graph paper.

The log of 2 to the base 10 is approximately 0.3. The distance from 1 ($\log_{10} 1 = 0$) to 2 is therefore 30% of the span. The log of 3 to the base 10 is 0.4771 or almost 48% of the span (very close to one-half the distance between power of 10 increments on the log scale). Since $\log_{10} 5 \cong 0.7$, it is marked off at a point 70% of the distance. Note that between any two digits the same compression of the lines appears as you progress from the left to the right. It is important to note the resulting numerical value and the spacing, since plots will typically only have the tic marks indicated in Fig. 11.2 due to a lack of space. You must realize that the longer bars for this figure have the numerical values of 0.3, 3, and 30 associated with them, whereas the next shorter bars have values of 0.5, 5, and 50 and the shortest bars 0.7, 7, and 70.

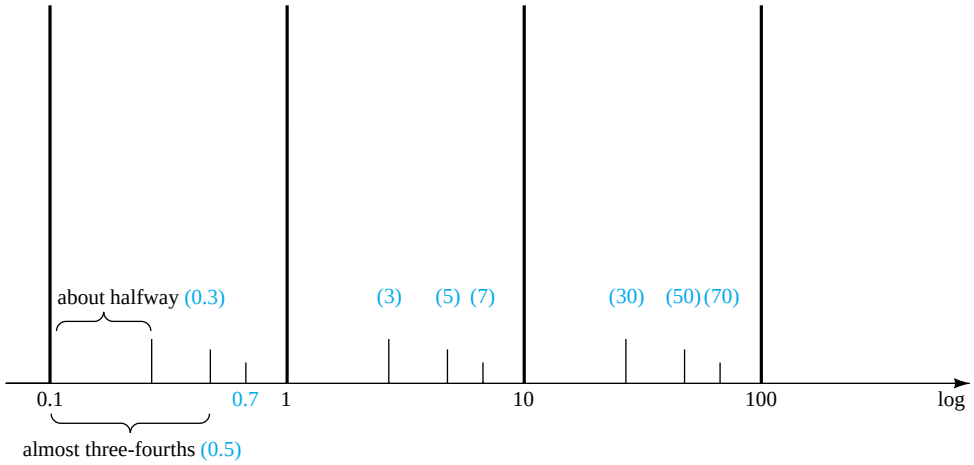


Figure 11.2 Identifying the numerical values of the tic marks on a log scale.

Be aware that plotting a function on a log scale can change the general appearance of the waveform as compared to a plot on a linear scale. A straight-line plot on a linear scale can develop a curve on a log scale, and a nonlinear plot on a linear scale can take on the appearance of a straight line on a log plot. The important point is that the results extracted at each level be correctly labeled by developing a familiarity with the spacing of Figs. 11.1 and 11.2. This is particularly true for some of the log-log plots that appear later in the book.

11.3 DECIBELS

The concept of the decibel (dB) and the associated calculations will become increasingly important in the remaining sections of this chapter. The background surrounding the term *decibel* has its origin in the established fact that power and audio levels are related on a logarithmic basis. That is, an increase in power level, say 4 to 16 W, does not result in an audio level increase by a factor of 16/4 = 4. It will increase by a factor of 2 as derived from the power of 4 in the following manner: $(4)^2 = 16$. For a change of 4 to 64 W, the audio level will increase by a factor of 3 since $(4)^3 = 64$. In logarithmic form, the relationship can be written as $\log_4 64 = 3$.

The term *bel* was derived from the surname of Alexander Graham Bell. For standardization, the bel (B) was defined by the following equation to relate power levels P_1 and P_2:

$$G = \log_{10} \frac{P_2}{P_1} \quad \text{bel} \tag{11.9}$$

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It was found, however, that the bel was too large a unit of measurement for practical purposes, so the decibel (dB) was defined such that 10 decibels = 1 bel. Therefore,

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} \quad \text{dB} \tag{11.10}$$

The terminal rating of electronic communication equipment (amplifiers, microphones, etc.) is commonly rated in decibels. Equation (11.10) indicates clearly, however, that the decibel rating is a measure of the difference in magnitude between *two* power levels. For a specified terminal (output) power (P_2) there must be a reference power level (P_1). The reference level is generally accepted to be 1 mW, although on occasion, the 6-mW standard of earlier years is applied. The resistance to be associated with the 1-mW power level is 600 Ω, chosen because it is the characteristic impedance of audio transmission lines. When the 1-mW level is employed as the reference level, the decibel symbol frequently appears as dBm. In equation form,

$$G_{\text{dBm}} = 10 \log_{10} \frac{P_2}{1 \text{ mW}}\bigg|_{600\ \Omega} \quad \text{dBm} \tag{11.11}$$

There exists a second equation for decibels that is applied frequently. It can be best described through the system of Fig. 11.3. For V_i equal to some value V_1, $P_1 = V_1^2/R_i$, where R_i, is the input resistance of the system of Fig. 11.3. If V_i should be increased (or decreased) to some other level, V_2, then $P_2 = V_2^2/R_i$. If we substitute into Eq. (11.10) to determine the resulting difference in decibels between the power levels,

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{V_2^2/R_i}{V_1^2/R_i} = 10 \log_{10} \left(\frac{V_2}{V_1}\right)^2$$

and

$$G_{\text{dB}} = 20 \log_{10} \frac{V_2}{V_1} \quad \text{dB} \tag{11.12}$$

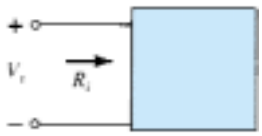


Figure 11.3 Configuration employed in the discussion of Eq. (11.12).

Frequently, the effect of different impedances ($R_1 \neq R_2$) is ignored and Eq. (11.12) applied simply to establish a basis of comparison between levels—voltage or current. For situations of this type, the decibel gain should more correctly be referred to as the voltage or current gain in decibels to differentiate it from the common usage of decibel as applied to power levels.

One of the advantages of the logarithmic relationship is the manner in which it can be applied to cascaded stages. For example, the magnitude of the overall voltage gain of a cascaded system is given by

$$|A_{v_T}| = |A_{v_1}||A_{v_2}||A_{v_3}|\cdots|A_{v_n}| \tag{11.13}$$

Applying the proper logarithmic relationship results in

$$G_v = 20 \log_{10} |A_{v_T}| = 20 \log_{10} |A_{v_1}| + 20 \log_{10} |A_{v_2}| + 20 \log_{10} |A_{v_3}| + \cdots + 20 \log_{10} |A_{v_n}| \quad \text{(dB)} \tag{11.14}$$

In words, the equation states that the decibel gain of a cascaded system is simply the sum of the decibel gains of each stage, that is,

$$\boxed{G_v = G_{v_1} + G_{v_2} + G_{v_3} + \cdots + G_{v_n}} \quad \text{dB} \tag{11.15}$$

In an effort to develop some association between dB levels and voltage gains, Table 11.2 was developed. First note that a gain of 2 results in a dB level of +6 dB while a drop to $\frac{1}{2}$ results in a −6-dB level. A change in V_o/V_i from 1 to 10, 10 to 100, or 100 to 1000 results in the same 20-dB change in level. When $V_o = V_i$, $V_o/V_i = 1$ and the dB level is 0. At a very high gain of 1000, the dB level is 60, while at the much higher gain of 10,000, the dB level is 80 dB, an increase of only 20 dB—a result of the logarithmic relationship. Table 11.2 clearly reveals that voltage gains of 50 dB or higher should immediately be recognized as being quite high.

TABLE 11.2

Voltage Gain, V_o/V_i	*dB Level*
0.5	−6
0.707	−3
1	0
2	6
10	20
40	32
100	40
1000	60
10,000	80
etc.	

EXAMPLE 11.5

Find the magnitude gain corresponding to a decibel gain of 100.

Solution

By Eq. (11.10),

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} = 100 \text{ dB} \rightarrow \log_{10} \frac{P_2}{P_1} = 10$$

so that

$$\frac{P_2}{P_1} = \mathbf{10^{10} = 10{,}000{,}000{,}000}$$

This example clearly demonstrates the range of decibel values to be expected from practical devices. Certainly, a future calculation giving a decibel result in the neighborhood of 100 should be questioned immediately.

EXAMPLE 11.6

The input power to a device is 10,000 W at a voltage of 1000 V. The output power is 500 W, while the output impedance is 20 Ω.

(a) Find the power gain in decibels.

(b) Find the voltage gain in decibels.

(c) Explain why parts (a) and (b) agree or disagree.

Solution

(a) $$G_{\text{dB}} = 10 \log_{10} \frac{P_o}{P_i} = 10 \log_{10} \frac{500 \text{ W}}{10 \text{ kW}} = 10 \log_{10} \frac{1}{20} = -10 \log_{10} 20$$

$$= -10(1.301) = \mathbf{-13.01 \text{ dB}}$$

(b) $$G_v = 20 \log_{10} \frac{V_o}{V_i} = 20 \log_{10} \frac{\sqrt{PR}}{1000} = 20 \log_{10} \frac{\sqrt{(500 \text{ W})(20 \ \Omega)}}{1000 \text{ V}}$$

$$= 20 \log_{10} \frac{100}{1000} = 20 \log_{10} \frac{1}{10} = -20 \log_{10} 10 = \mathbf{-20 \text{ dB}}$$

(c) $$R_i = \frac{V_i^2}{P_i} = \frac{(1 \text{ kV})^2}{10 \text{ kW}} = \frac{10^6}{10^4} = \mathbf{100 \ \Omega} \neq R_o = \mathbf{20 \ \Omega}$$

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EXAMPLE 11.7

An amplifier rated at 40-W output is connected to a 10-Ω speaker.

(a) Calculate the input power required for full power output if the power gain is 25 dB.

(b) Calculate the input voltage for rated output if the amplifier voltage gain is 40 dB.

Solution

(a) Eq. (11.10): $25 = 10 \log_{10} \frac{40\text{ W}}{P_i} \Rightarrow P_i = \frac{40\text{ W}}{\text{antilog } (2.5)} = \frac{40\text{ W}}{3.16 \times 10^2}$

$$= \frac{40\text{ W}}{316} \cong \mathbf{126.5\ mW}$$

(b) $G_v = 20 \log_{10} \frac{V_o}{V_i} \Rightarrow 40 = 20 \log_{10} \frac{V_o}{V_i}$

$$\frac{V_o}{V_i} = \text{antilog } 2 = 100$$

$$V_o = \sqrt{PR} = \sqrt{(40\text{ W})(10\text{ V})} = 20\text{ V}$$

$$V_i = \frac{V_o}{100} = \frac{20\text{ V}}{100} = 0.2\text{ V} = \mathbf{200\ mV}$$

11.4 GENERAL FREQUENCY CONSIDERATIONS

The frequency of the applied signal can have a pronounced effect on the response of a single-stage or multistage network. The analysis thus far has been for the midfrequency spectrum. At low frequencies, we shall find that the coupling and bypass capacitors can no longer be replaced by the short-circuit approximation because of the increase in reactance of these elements. The frequency-dependent parameters of the small-signal equivalent circuits and the stray capacitive elements associated with the active device and the network will limit the high-frequency response of the system. An increase in the number of stages of a cascaded system will also limit both the high- and low-frequency responses.

The magnitudes of the gain response curves of an *RC*-coupled, direct-coupled, and transformer-coupled amplifier system are provided in Fig. 11.4. Note that the horizontal scale is a logarithmic scale to permit a plot extending from the low- to the high-frequency regions. For each plot, a low-, high-, and mid-frequency region has been defined. In addition, the primary reasons for the drop in gain at low and high frequencies have also been indicated within the parentheses. For the *RC*-coupled amplifier, the drop at low frequencies is due to the increasing reactance of C_C, C_s, or C_E, while its upper frequency limit is determined by either the parasitic capacitive elements of the network and frequency dependence of the gain of the active device. An explanation of the drop in gain for the transformer-coupled system requires a basic understanding of "transformer action" and the transformer equivalent circuit. For the moment, let us say that it is simply due to the "shorting effect" (across the input terminals of the transformer) of the magnetizing inductive reactance at low frequencies ($X_L = 2\pi fL$). The gain must obviously be zero at $f = 0$ since at this point there is no longer a changing flux established through the core to induce a secondary or output voltage. As indicated in Fig. 11.4, the high-frequency response is controlled primarily by the stray capacitance between the turns of the primary and secondary wind-

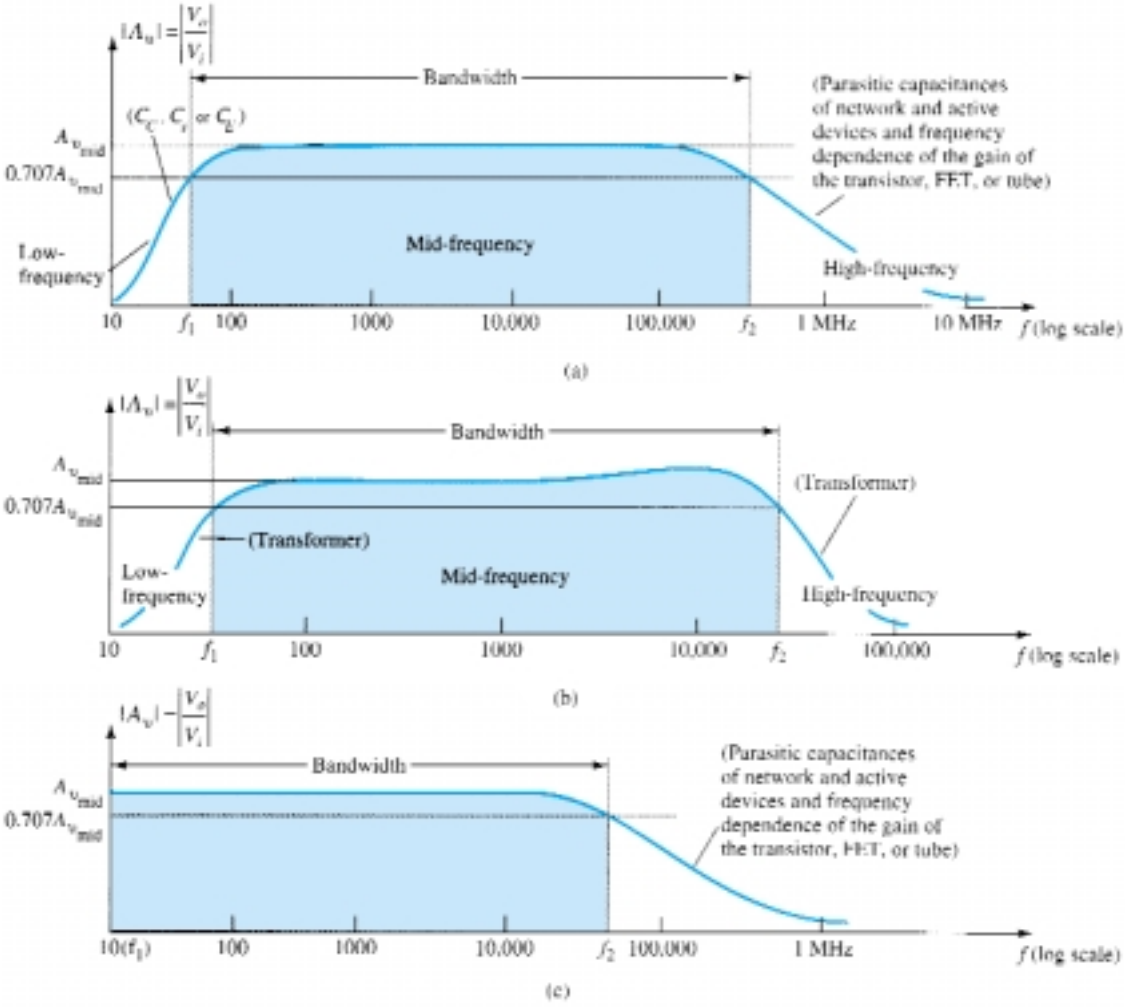


Figure 11.4 Gain versus frequency: (a) *RC*-coupled amplifiers; (b) transformer-coupled amplifiers; (c) direct-coupled amplifiers.

ings. For the direct-coupled amplifier, there are no coupling or bypass capacitors to cause a drop in gain at low frequencies. As the figure indicates, it is a flat response to the upper cutoff frequency, which is determined by either the parasitic capacitances of the circuit or the frequency dependence of the gain of the active device.

For each system of Fig. 11.4, there is a band of frequencies in which the magnitude of the gain is either equal or relatively close to the midband value. To fix the frequency boundaries of relatively high gain, $0.707A_{v_{\text{mid}}}$ was chosen to be the gain at the cutoff levels. The corresponding frequencies f_1 and f_2 are generally called the *corner, cutoff, band, break,* or *half-power frequencies.* The multiplier 0.707 was chosen because at this level the output power is half the midband power output, that is, at midfrequencies,

$$P_{o_{\text{mid}}} = \frac{|V^2_o|}{R_o} = \frac{|A_{v_{\text{mid}}} V_i|^2}{R_o}$$

and at the half-power frequencies,

$$P_{o_{HPF}} = \frac{|0.707A_{v_{\text{mid}}} V_i|^2}{R_o} = 0.5\,\frac{|A_{v_{\text{mid}}} V_i|^2}{R_o}$$

and $$P_{o_{HPF}} = 0.5P_{o_{mid}} \tag{11.16}$$

The bandwidth (or passband) of each system is determined by f_1 and f_2, that is,

$$\text{bandwidth (BW)} = f_2 - f_1 \tag{11.17}$$

For applications of a communications nature (audio, video), a decibel plot of the voltage gain versus frequency is more useful than that appearing in Fig. 11.4. Before obtaining the logarithmic plot, however, the curve is generally normalized as shown in Fig. 11.5. In this figure, the gain at each frequency is divided by the midband value. Obviously, the midband value is then 1 as indicated. At the half-power frequencies, the resulting level is $0.707 = 1/\sqrt{2}$. A decibel plot can now be obtained by applying Eq. (11.12) in the following manner:

$$\frac{A_v}{A_{v_{mid}}}\Big|_{dB} = 20 \log_{10} \frac{A_v}{A_{v_{mid}}} \tag{11.18}$$

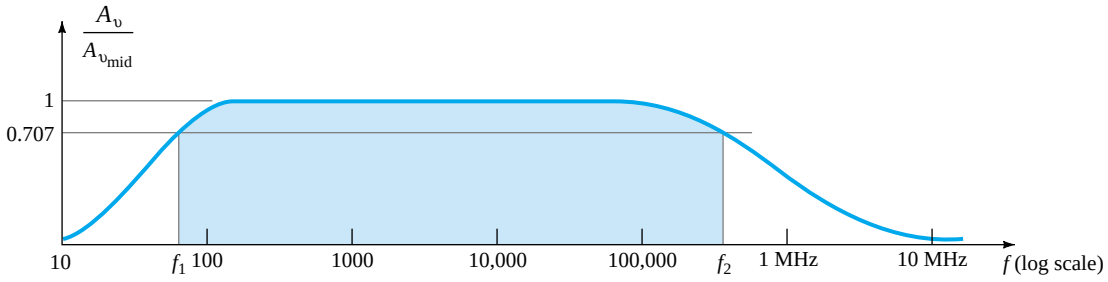


Figure 11.5 Normalized gain versus frequency plot.

At midband frequencies, $20 \log_{10} 1 = 0$, and at the cutoff frequencies, $20 \log_{10} 1/\sqrt{2} = -3$ dB. Both values are clearly indicated in the resulting decibel plot of Fig. 11.6. The smaller the fraction ratio, the more negative the decibel level.

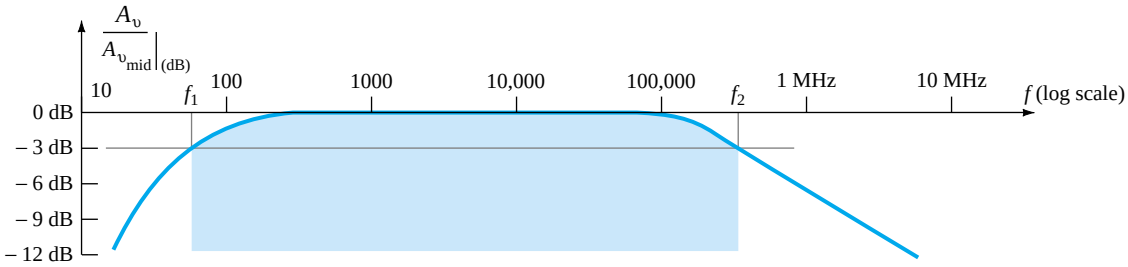


Figure 11.6 Decibel plot of the normalized gain versus frequency plot of Fig. 11.5.

For the greater part of the discussion to follow, a decibel plot will be made only for the low- and high-frequency regions. Keep Fig. 11.6 in mind, therefore, to permit a visualization of the broad system response.

It should be understood that most amplifiers introduce a 180° phase shift between input and output signals. This fact must now be expanded to indicate that this is the case only in the midband region. At low frequencies, there is a phase shift such that V_o lags V_i by an increased angle. At high frequencies, the phase shift will drop below 180°. Figure 11.7 is a standard phase plot for an *RC*-coupled amplifier.

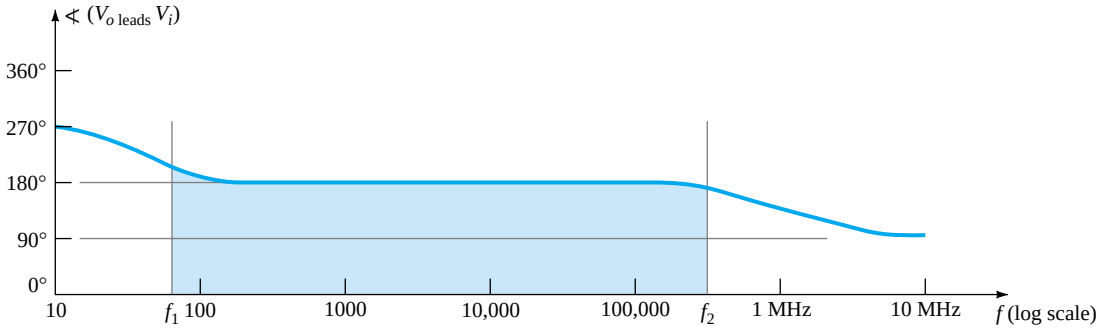


Figure 11.7 Phase plot for an *RC*-coupled amplifier system.

11.5 LOW-FREQUENCY ANALYSIS—BODE PLOT

In the low-frequency region of the single-stage BJT or FET amplifier, it is the *R-C* combinations formed by the network capacitors C_C, C_E, and C_s and the network resistive parameters that determine the cutoff frequencies. In fact, an *R-C* network similar to Fig. 11.8 can be established for each capacitive element and the frequency at which the output voltage drops to 0.707 of its maximum value determined. Once the cutoff frequencies due to each capacitor are determined, they can be compared to establish which will determine the low-cutoff frequency for the system.

Our analysis, therefore, will begin with the series *R-C* combination of Fig. 11.8 and the development of a procedure that will result in a plot of the frequency response with a minimum of time and effort. At very high frequencies,

$$X_C = \frac{1}{2\pi fC} \cong 0\ \Omega$$

and the short-circuit equivalent can be substituted for the capacitor as shown in Fig. 11.9. The result is that $V_o \cong V_i$ at high frequencies. At $f = 0$ Hz,

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(0)C} = \infty\ \Omega$$

and the open-circuit approximation can be applied as shown in Fig. 11.10, with the result that $V_o = 0$ V.

Between the two extremes, the ratio $A_v = V_o/V_i$ will vary as shown in Fig. 11.11. As the frequency increases, the capacitive reactance decreases and more of the input voltage appears across the output terminals.

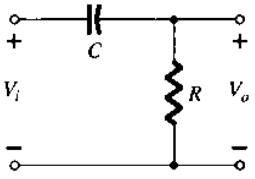


Figure 11.8 *R-C* combination that will define a low cutoff frequency.

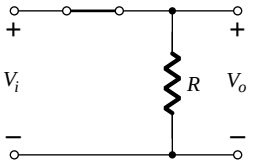


Figure 11.9 *R-C* circuit of Figure 11.8 at very high frequencies.

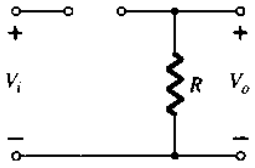


Figure 11.10 *R-C* circuit of Figure 11.8 at $f = 0$ Hz.

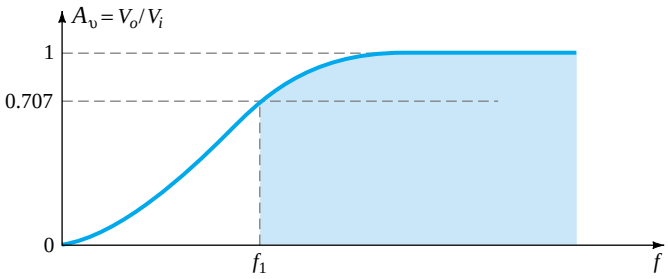


Figure 11.11 Low frequency response for the *R-C* circuit of Figure 11.8.

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The output and input voltages are related by the voltage-divider rule in the following manner:

$$\mathbf{V}_o = \frac{\mathbf{R}\mathbf{V}_i}{\mathbf{R} + \mathbf{X}_C}$$

with the magnitude of V_o determined by

$$V_o = \frac{RV_i}{\sqrt{R^2 + X_C^2}}$$

For the special case where $X_C = R$,

$$V_o = \frac{RV_i}{\sqrt{R^2X_C^2}} = \frac{RV_i}{} = \frac{RV_i}{\sqrt{2R^2}} = \frac{RV_i}{\sqrt{2R}} = \frac{1}{\sqrt{2}}\,V_i$$

and

$$\boxed{|A_v| = \frac{V_o}{V_i} = \frac{1}{\sqrt{2}} = 0.707|_{X_C=R}} \tag{11.19}$$

the level of which is indicated on Fig. 11.11. In other words, at the frequency of which $X_C = R$, the output will be 70.7% of the input for the network of Fig. 11.8.

The frequency at which this occurs is determined from

$$X_C = \frac{1}{2\pi f_1 C} = R$$

and

$$\boxed{f_1 = \frac{1}{2\pi RC}} \tag{11.20}$$

In terms of logs,

$$G_v = 20\log_{10} A_v = 20\log_{10}\frac{1}{\sqrt{2}} = -3 \text{ dB}$$

while at $A_v = V_o/V_i = 1$ or $V_o = V_i$ (the maximum value),

$$G_v = 20\log_{10} 1 = 20(0) = 0 \text{ dB}$$

In Fig. 11.6, we recognize that there is a 3-dB drop in gain from the midband level when $f = f_1$. In a moment, we will find that an *RC* network will determine the low-frequency cutoff frequency for a BJT transistor and f_1 will be determined by Eq. (11.20).

If the gain equation is written as

$$A_v = \frac{V_o}{V_i} = \frac{R}{R - jX_C} = \frac{1}{1 - j(X_C/R)} = \frac{1}{1 - j(1/\omega CR)} = \frac{1}{1 - j(1/2\pi fCR)}$$

and using the frequency defined above,

$$\boxed{A_v = \frac{1}{1 - j(f_1/f)}} \tag{11.21}$$

In the magnitude and phase form,

$$A_v = \frac{V_o}{V_i} = \underbrace{\frac{1}{\sqrt{1 + (f_1/f)^2}}}_{\text{magnitude of } A_v}\ \underbrace{\angle \tan^{-1}(f_1/f)}_{\substack{\text{phase } \measuredangle \text{ by which} \\ V_o \text{ leads } V_i}} \tag{11.22}$$

For the magnitude when $f = f_1$,

$$|A_v| = \frac{1}{\sqrt{1 + (1)^2}} = \frac{1}{\sqrt{2}} = 0.707 \rightarrow -3 \text{ dB}$$

In the logarithmic form, the gain in dB is

$$A_{v(\text{dB})} = 20 \log_{10} \frac{1}{\sqrt{1 + (f_1/f)^2}} = -20 \log_{10}\left[1 + \left(\frac{f_1}{f}\right)^2\right]^{1/2}$$

$$= -(\tfrac{1}{2})(20) \log_{10}\left[1 + \left(\frac{f_1}{f}\right)^2\right]$$

$$= -10 \log_{10}\left[1 + \left(\frac{f_1}{f}\right)^2\right]$$

For frequencies where $f \ll f_1$ or $(f_1/f)^2 \gg 1$, the equation above can be approximated by

$$A_{v(\text{dB})} = -10 \log_{10}\left(\frac{f_1}{f}\right)^2$$

and finally,

$$\boxed{A_{v(\text{dB})} = -20 \log_{10} \frac{f_1}{f}} \qquad f \ll f_1 \qquad (11.23)$$

Ignoring the condition $f \ll f_1$ for a moment, a plot of Eq. (11.23) on a frequency log scale will yield a result of a very useful nature for future decibel plots.

At $f = f_1$: $\dfrac{f_1}{f} = 1$ and $-20 \log_{10} 1 = 0$ dB

At $f = \frac{1}{2} f_1$: $\dfrac{f_1}{f} = 2$ and $-20 \log_{10} 2 \cong -6$ dB

At $f = \frac{1}{4} f_1$: $\dfrac{f_1}{f} = 4$ and $-20 \log_{10} 4 \cong -12$ dB

At $f = \frac{1}{10} f_1$: $\dfrac{f_1}{f} = 10$ and $-20 \log_{10} 10 = -20$ dB

A plot of these points is indicated in Fig. 11.12 from $0.1f_1$ to f_1. Note that this results in a straight line when plotted against a log scale. In the same figure, a straight

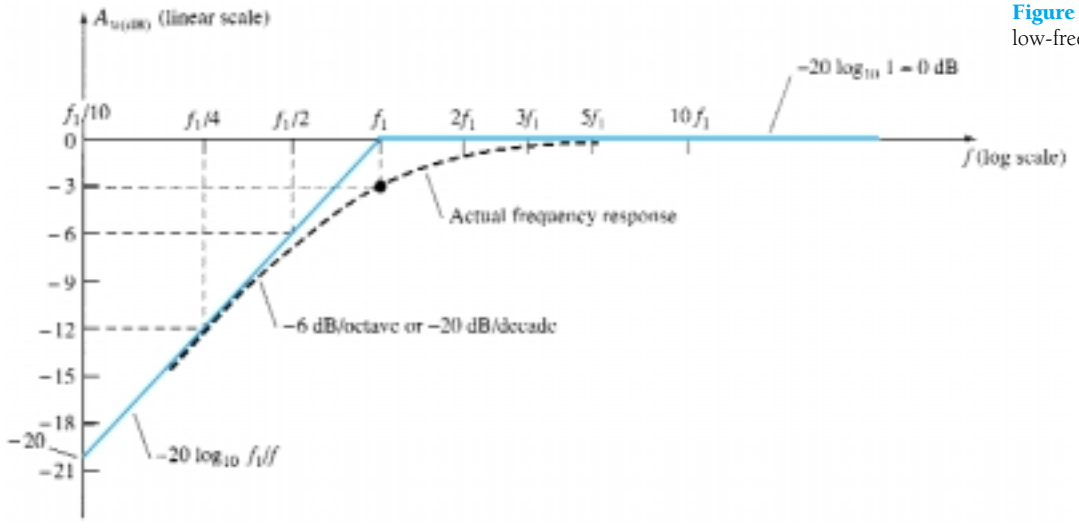


Figure 11.12 Bode plot for the low-frequency region.

line is also drawn for the condition of 0 dB for $f \gg f_1$. As stated earlier, the straight-line segments (asymptotes) are only accurate for 0 dB when $f \gg f_1$ and the sloped line when $f_1 \gg f$. We know, however, that when $f = f_1$, there is a 3-dB drop from the mid-band level. Employing this information in association with the straight-line segments permits a fairly accurate plot of the frequency response as indicated in the same figure. The piecewise linear plot of the asymptotes and associated breakpoints is called a *Bode plot* of the magnitude versus frequency.

The calculations above and the curve itself demonstrate clearly that:

A change in frequency by a factor of 2, equivalent to 1 octave, results in a 6-dB change in the ratio as noted by the change in gain from $f_1/2$ to f_1.

As noted by the change in gain from $f_1/2$ to f_1:

For a 10:1 change in frequency, equivalent to 1 decade, there is a 20-dB change in the ratio as demonstrated between the frequencies of $f_1/10$ and f_1.

In the future, therefore, a decibel plot can easily be obtained for a function having the format of Eq. (11.23). First, simply find f_1 from the circuit parameters and then sketch two asymptotes—one along the 0-dB line and the other drawn through f_1 sloped at 6 dB/octave or 20 dB/decade. Then, find the 3-dB point corresponding to f_1 and sketch the curve.

EXAMPLE 11.8

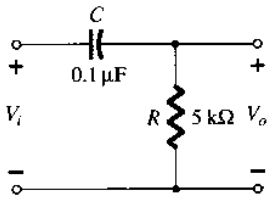


Figure 11.13 Example 11.8

For the network of Fig. 11.13:

(a) Determine the break frequency.

(b) Sketch the asymptotes and locate the −3-dB point.

(c) Sketch the frequency response curve.

Solution

(a) $$f_1 = \frac{1}{2\pi RC} = \frac{1}{(6.28)(5 \times 10^3\ \Omega)(0.1 \times 10^{-6}\ \text{F})}$$

$$\cong \mathbf{318.5\ Hz}$$

(b) and (c). See Fig. 11.14.

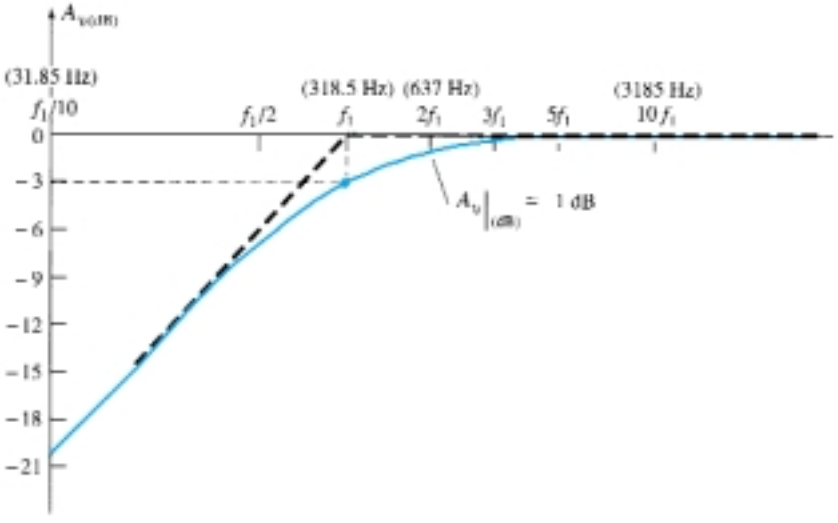


Figure 11.14 Frequency response for the *R-C* circuit of Figure 11.13.

The gain at any frequency can then be determined from the frequency plot in the following manner:

$$A_{v(\text{dB})} = 20 \log_{10} \frac{V_o}{V_i}$$

but

$$\frac{A_{v(\text{dB})}}{20} = \log_{10} \frac{V_o}{V_i}$$

and

$$\boxed{A_v = \frac{V_o}{V_i} = 10^{\left(\frac{A_{v(\text{dB})}}{20}\right)}} \tag{11.24}$$

For example, if $A_{v(\text{dB})} = -3$ dB,

$$A_v = \frac{V_o}{V_i} = 10^{(-3/20)} = 10^{(-0.15)} \cong 0.707 \qquad \text{as expected}$$

The quantity $10^{-0.15}$ is determined using the 10^x function found on most scientific calculators.

From Fig. 11.14, $A_{v(\text{dB})} \cong -1$ dB at $f = 2f_1 = 637$ Hz. The gain at this point is

$$A_v = \frac{V_o}{V_i} = 10^{\left(\frac{A_{v(\text{dB})}}{20}\right)} = 10^{(-1/20)} = 10^{(-0.05)} = 0.891$$

and

$$V_o = 0.891 V_i$$

or V_o is 89.1% of V_i at $f = 637$ Hz.

The phase angle of θ is determined from

$$\boxed{\theta = \tan^{-1} \frac{f_1}{f}} \tag{11.25}$$

from Eq. (11.22).

For frequencies $f \ll f_1$,

$$\theta = \tan^{-1} \frac{f_1}{f} \rightarrow 90°$$

For instance, if $f_1 = 100f$,

$$\theta = \tan^{-1} \frac{f_1}{f} = \tan^{-1}(100) = 89.4°$$

For $f = f_1$,

$$\theta = \tan^{-1} \frac{f_1}{f} = \tan^{-1} 1 = 45°$$

For $f \gg f_1$,

$$\theta = \tan^{-1} \frac{f_1}{f} \rightarrow 0°$$

For instance, if $f = 100f_1$,

$$\theta = \tan^{-1} \frac{f_1}{f} = \tan^{-1} 0.01 = 0.573°$$

f

A plot of $\theta = \tan^{-1}(f_1/f)$ is provided in Fig. 11.15. If we add the additional 180° phase shift introduced by an amplifier, the phase plot of Fig. 11.7 will be obtained. The magnitude and phase response for an *R-C* combination have now been established. In Section 11.6, each capacitor of importance in the low-frequency region will be redrawn in an *R-C* format and the cutoff frequency for each determined to establish the low-frequency response for the BJT amplifier.

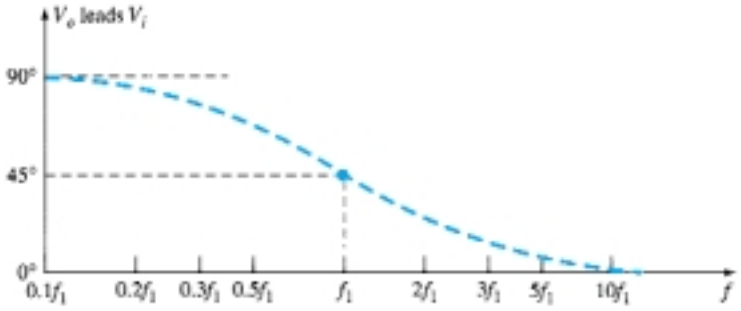


Figure 11.15 Phase response for the *R-C* circuit of Figure 11.8.

11.6 LOW-FREQUENCY RESPONSE — BJT AMPLIFIER

The analysis of this section will employ the loaded voltage-divider BJT bias configuration, but the results can be applied to any BJT configuration. It will simply be necessary to find the appropriate equivalent resistance for the *R-C* combination. For the network of Fig. 11.16, the capacitors C_s, C_C, and C_E will determine the low-frequency response. We will now examine the impact of each independently in the order listed.

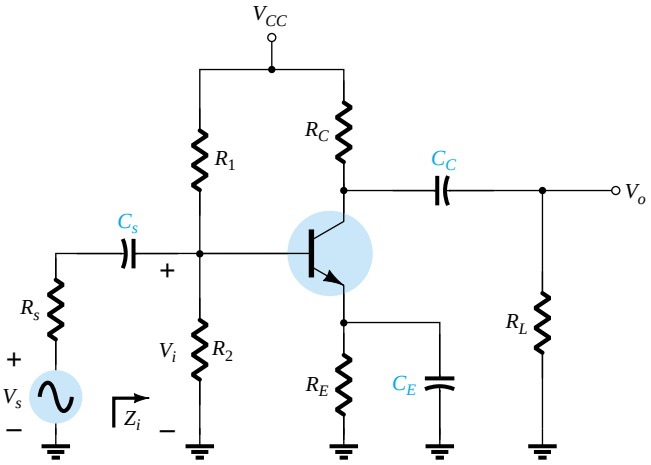


Figure 11.16 Loaded BJT amplifier with capacitors that affect the low-frequency response.

C_s

Since C_s is normally connected between the applied source and the active device, the general form of the *R-C* configuration is established by the network of Fig. 11.17. The total resistance is now $R_s + R_i$, and the cutoff frequency as established in Section 11.5 is

$$f_{L_s} = \frac{1}{2\pi\,(R_s + R_i)C_s} \tag{11.26}$$

At mid or high frequencies, the reactance of the capacitor will be sufficiently small to permit a short-circuit approximation for the element. The voltage V_i will then be related to V_s by

$$V_i\big|_{\text{mid}} = \frac{R_i V_s}{R_i + R_s} \tag{11.27}$$

At f_{L_s}, the voltage V_i will be 70.7% of the value determined by Eq. (11.27), assuming that C_s is the only capacitive element controlling the low-frequency response.

For the network of Fig. 11.16, when we analyze the effects of C_s we must make the assumption that C_E and C_C are performing their designed function or the analysis becomes too unwieldy, that is, that the magnitude of the reactances of C_E and C_C permits employing a short-circuit equivalent in comparison to the magnitude of the other series impedances. Using this hypothesis, the ac equivalent network for the input section of Fig. 11.16 will appear as shown in Fig. 11.18.

The value of R_i for Eq. (11.26) is determined by

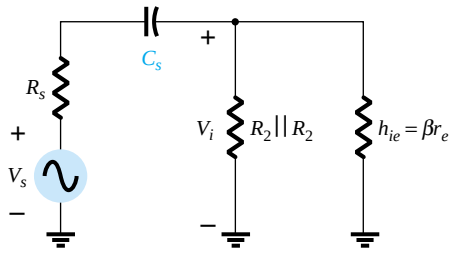


Figure 11.18 Localized ac equivalent for C_s.

$$R_i = R_1 \| R_2 \| \beta r_e \tag{11.28}$$

The voltage $\mathbf{V}_i$ applied to the input of the active device can be calculated using the voltage-divider rule:

$$\mathbf{V}_i = \frac{R_i \mathbf{V}_s}{R_s + R_i - jX_{C_s}} \tag{11.29}$$

C_C

Since the coupling capacitor is normally connected between the output of the active device and the applied load, the *R-C* configuration that determines the low cutoff frequency due to C_C appears in Fig. 11.19. From Fig. 11.19, the total series resistance is now $R_o + R_L$ and the cutoff frequency due to C_C is determined by

$$f_{L_C} = \frac{1}{2\pi\,(R_o + R_L)C_C} \tag{11.30}$$

Ignoring the effects of C_s and C_E, the output voltage V_o will be 70.7% of its midband value at f_{L_C}. For the network of Fig. 11.16, the ac equivalent network for the output section with $V_i = 0$ V appears in Fig. 11.20. The resulting value for R_o in Eq. (11.30) is then simply

$$R_o = R_C \| r_o \tag{11.31}$$

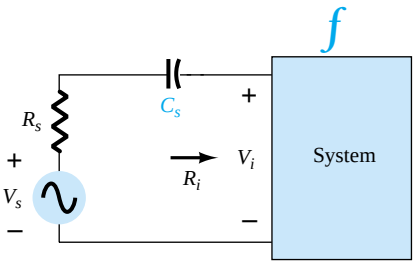


Figure 11.17 Determining the effect of C_s on the low frequency response.

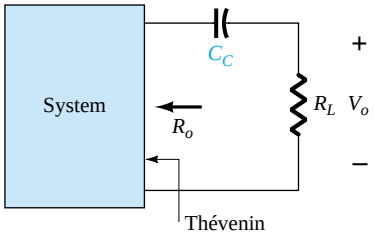


Figure 11.19 Determining the effect of C_C on the low-frequency response.

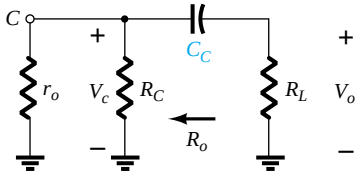


Figure 11.20 Localized ac equivalent for C_C with $V_i = 0$ V.

f

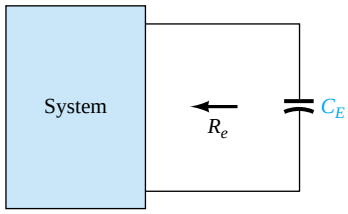


Figure 11.21 Determining the effect of C_E on the low-frequency response.

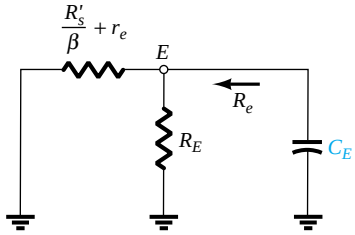


Figure 11.22 Localized ac equivalent of C_E.

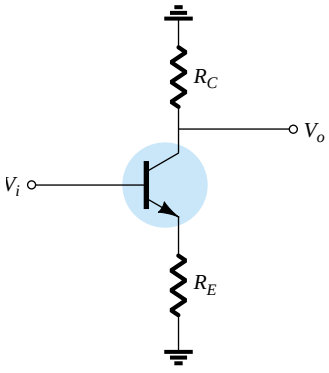


Figure 11.23 Network employed to describe the effect of C_E on the amplifier gain.

C_E

To determine f_{L_E}, the network "seen" by C_E must be determined as shown in Fig. 11.21. Once the level of R_e is established, the cutoff frequency due to C_E can be determined using the following equation:

$$f_{L_E} = \frac{1}{2\pi R_e C_E} \tag{11.32}$$

For the network of Fig. 11.16, the ac equivalent as "seen" by C_E appears in Fig. 11.22. The value of R_e is therefore determined by

$$R_e = R_E \| \left(\frac{R'_s}{\beta} + r_e \right) \tag{11.33}$$

where $R'_s = R_s \| R_1 \| R_2$.

The effect of C_E on the gain is best described in a quantitative manner by recalling that the gain for the configuration of Fig. 11.23 is given by

$$A_v = \frac{-R_C}{r_e + R_E}$$

The maximum gain is obviously available where R_E is zero ohms. At low frequencies, with the bypass capacitor C_E in its "open-circuit" equivalent state, all of R_E appears in the gain equation above, resulting in the minimum gain. As the frequency increases, the reactance of the capacitor C_E will decrease, reducing the parallel impedance of R_E and C_E until the resistor R_E is effectively "shorted out" by C_E. The result is a maximum or midband gain determined by $A_v = -R_C/r_e$. At f_{L_E} the gain will be 3 dB below the midband value determined with R_E "shorted out."

Before continuing, keep in mind that C_s, C_C, and C_E will affect only the low-frequency response. At the midband frequency level, the short-circuit equivalents for the capacitors can be inserted. Although each will affect the gain $A_v = V_o/V_i$ in a similar frequency range, the highest low-frequency cutoff determined by C_s, C_C, or C_E will have the greatest impact since it will be the last encountered before the midband level. If the frequencies are relatively far apart, the highest cutoff frequency will essentially determine the lower cutoff frequency for the entire system. If there are two or more "high" cutoff frequencies, the effect will be to raise the lower cutoff frequency and reduce the resulting bandwidth of the system. In other words, there is an interaction between capacitive elements that can affect the resulting low cutoff frequency. However, if the cutoff frequencies established by each capacitor are sufficiently separated, the effect of one on the other can be ignored with a high degree of accuracy—a fact that will be demonstrated by the printouts to appear in the following example.

EXAMPLE 11.9

(a) Determine the lower cutoff frequency for the network of Fig. 11.16 using the following parameters:

$C_s = 10\ \mu\text{F}$, $C_E = 20\ \mu\text{F}$, $C_C = 1\ \mu\text{F}$

$R_s = 1\ \text{k}\Omega$, $R_1 = 40\ \text{k}\Omega$, $R_2 = 10\ \text{k}\Omega$, $R_E = 2\ \text{k}\Omega$, $R_C = 4\ \text{k}\Omega$, $R_L = 2.2\ \text{k}\Omega$

$\beta = 100$, $r_o = \infty\ \Omega$, $V_{CC} = 20\ \text{V}$

(b) Sketch the frequency response using a Bode plot.

Solution

(a) Determining r_e for dc conditions:

$$\beta R_E = (100)(2\ \text{k}\Omega) = 200\ \text{k}\Omega \gg 10R_2 = 100\ \text{k}\Omega$$

The result is:

$$V_B \cong \frac{R_2 V_{CC}}{R_2 + R_1} = \frac{10\ \text{k}\Omega(20\ \text{V})}{10\ \text{k}\Omega + 40\ \text{k}\Omega} = \frac{200\ \text{V}}{50} = 4\ \text{V}$$

with

$$I_E = \frac{V_E}{R_E} = \frac{4\ \text{V} - 0.7\ \text{V}}{2\ \text{k}\Omega} = \frac{3.3\ \text{V}}{2\ \text{k}\Omega} = 1.65\ \text{mA}$$

so that

$$r_e = \frac{26\ \text{mV}}{1.65\ \text{mA}} \cong \mathbf{15.76\ \Omega}$$

and

$$\beta r_e = 100(15.76\ \Omega) = 1576\ \Omega = \mathbf{1.576\ k\Omega}$$

Midband Gain

$$A_v = \frac{V_o}{V_i} = \frac{-R_C \| R_L}{r_e} = -\frac{(4\ \text{k}\Omega)\|(2.2\ \text{k}\Omega)}{15.76\ \Omega} \cong -90$$

The input impedance

$$\begin{aligned} Z_i = R_i &= R_1 \| R_2 \| \beta r_e \\ &= 40\ \text{k}\Omega \| 10\ \text{k}\Omega \| 1.576\ \text{k}\Omega \\ &\cong 1.32\ \text{k}\Omega \end{aligned}$$

and from Fig. 11.24,

$$V_i = \frac{R_i V_s}{R_i + R_s}$$

or

$$\frac{V_i}{V_s} = \frac{R_i}{R_i + R_s} = \frac{1.32\ \text{k}\Omega}{1.32\ \text{k}\Omega + 1\ \text{k}\Omega} = 0.569$$

so that

$$\begin{aligned} A_{v_s} = \frac{V_o}{V_s} &= \frac{V_o}{V_i}\frac{V_i}{V_s} = (-90)(0.569) \\ &= \mathbf{-51.21} \end{aligned}$$

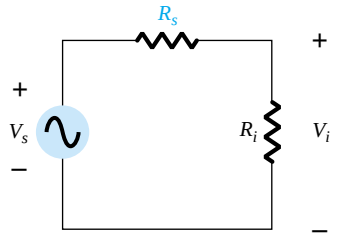


Figure 11.24 Determining the effect of R_s on the gain A_{v_s}.

C_s

$$R_i = R_1 \| R_2 \| \beta r_e = 40\ \text{k}\Omega \| 10\ \text{k}\Omega \| 1.576\ \text{k}\Omega \cong 1.32\ \text{k}\Omega$$

$$f_{L_s} = \frac{1}{2\pi (R_s + R_i) C_s} = \frac{1}{(6.28)(1\ \text{k}\Omega + 1.32\ \text{k}\Omega)(10\ \mu\text{F})}$$

$$f_{L_s} \cong \mathbf{6.86\ Hz}$$

f

The results just obtained will now be verified using PSpice Windows. The network with its various capacitors appears in Fig. 11.25. The **Model Editor** was used to set I_s to 2E-15A and beta to 100. The remaining parameters were removed from the listing to idealize the response to the degree possible. Under **Analysis Setup-AC Sweep,** the frequency was set to 10 kHz to establish a frequency in the midband region. A simulation of the network resulted in the dc levels of Fig. 11.25. Note that V_B is 3.9 V versus the calculated level of 4 V and that V_E is 3.2 V versus the calculated level of 3.3 V. Very close when you consider that the approximate model was used. V_{BE} is very close to the 0.7 V at 0.71 V. The output file reveals that the ac voltage across the load at a frequency of 10 kHz is 49.67 mV, resulting in a gain of 49.67, which is very close to the calculated level of 51.21.

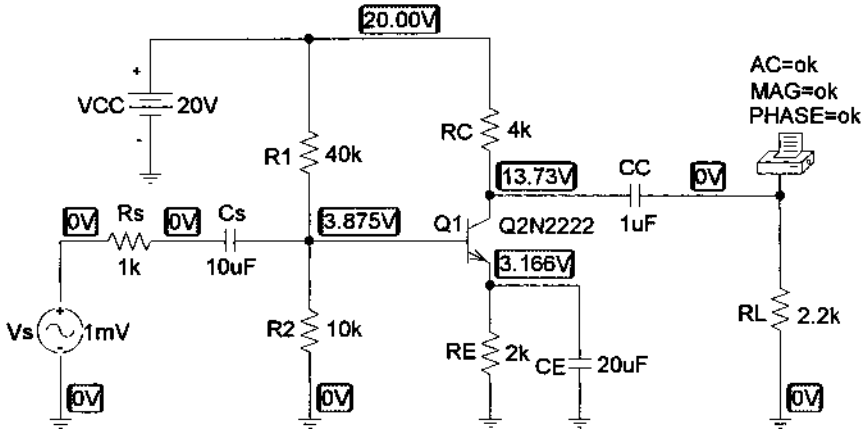


Figure 11.25 Network of Figure 11.16 with assigned values.

A plot of the gain versus frequency will now be obtained with only C_S as a determining factor. The other capacitors, C_C and C_E, will be set to very high values so they are essentially short circuits at any of the frequencies of interest. Setting C_C and C_E to 1 F will remove any affect they will have on the response on the low-frequency region. Here, one must be careful as the program does not recognize 1F as one Farad. It must be entered as 1E6uF. Since the pattern desired is gain versus frequency, we must use the sequence **Analysis-Setup-Analysis Setup-Enable AC Sweep-AC Sweep** to obtain the **AC Sweep and Noise Analysis** dialog box. Since our interest will be in the low-frequency range, we will choose a range of 1 Hz (0 Hz is an invalid entry) to 100 Hz. If you want a frequency range starting close to 0 Hz, you would have to choose a frequency such as 0.001 Hz or something small enough not to be noticeable on the plot. The **Total Pts.:** will be set at 1000 for a good continuous plot, the **Start Freq.:** at 1 Hz, and the **End Freq.:** at 100 Hz. The **AC Sweep Type** will be left on **Linear.** A simulation followed by **Trace-Add-V(RL:1)** will result in the desired plot. However, the computer has selected a log scale for the horizontal axis that extends from 1 Hz to 1 kHz even though we requested a linear scale. If we choose **Plot-X-Axis Settings-Linear-OK,** we will get a linear plot to 120 Hz, but the curve of interest is all in the low end—the log axis obviously provided a better plot for our region of interest. Returning to **Plot-X-Axis Settings** and choosing **Log,** we return to the original plot. Our interest only lies in the region of 1 to 100 Hz, so the remaining frequencies to 1 kHz should be removed with **Plot-X-Axis Settings-User Defined-1Hz to 100Hz-OK.** The vertical axis also goes to 60 mV, and we want to limit to 50 mV for this frequency range. This is accomplished with **Plot-Y-Axis Settings-User Defined-0V to 50mV-OK,** after which the pattern of Fig. 11.26 will be obtained.

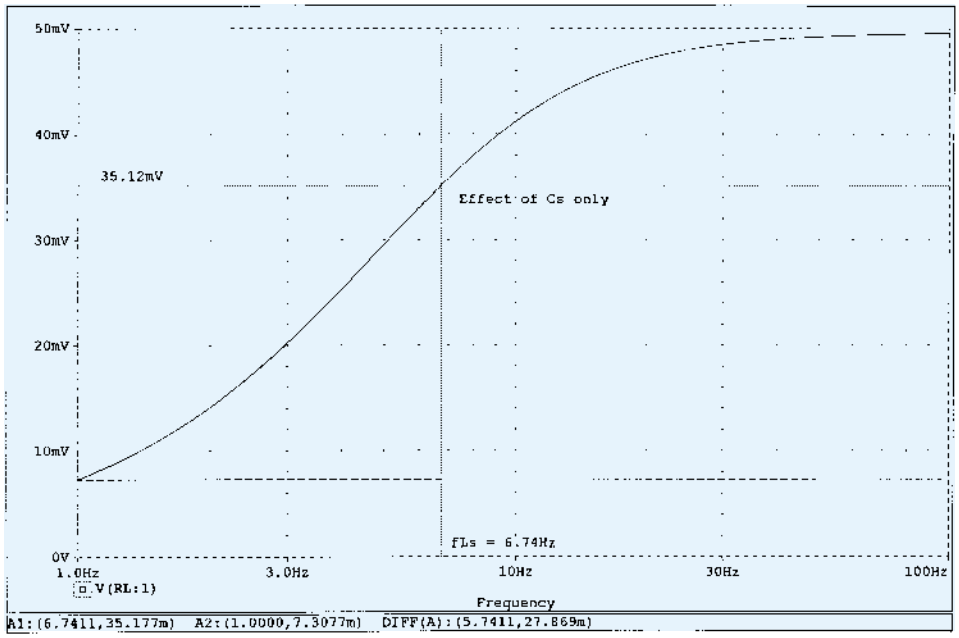


Figure 11.26 Low-frequency response due to C_S.

Note how closely the curve approaches 50 mV in this range. The cutoff level is determined by 0.707(49.67 mV) = 35.12 mV, which can be found by clicking the **Toggle cursor** icon and moving the intersection up the graph until the 35.177-mV level is reached for **A1.** At this point, the frequency of the horizontal axis can be read as 6.74 Hz, comparing very well to the predicted value of 6.86 Hz. Note that **A2** remains at the lowest level of the plot, at 1 Hz.

C_C

$$f_{L_C} = \frac{1}{2\pi(R_C + R_L)C_C}$$

$$= \frac{1}{(6.28)(4\ \text{k}\Omega + 2.2\ \text{k}\Omega)(1\ \mu\text{F})}$$

$$\cong \mathbf{25.68\ Hz}$$

To investigate the effects of C_C on the lower cutoff frequency, both C_S and C_E must be set to 1 Farad as described above. Following the procedure outlined above will result in the plot of Fig. 11.27, with a cutoff frequency of 25.58 Hz, providing a close match with the calculated level of 25.68 Hz.

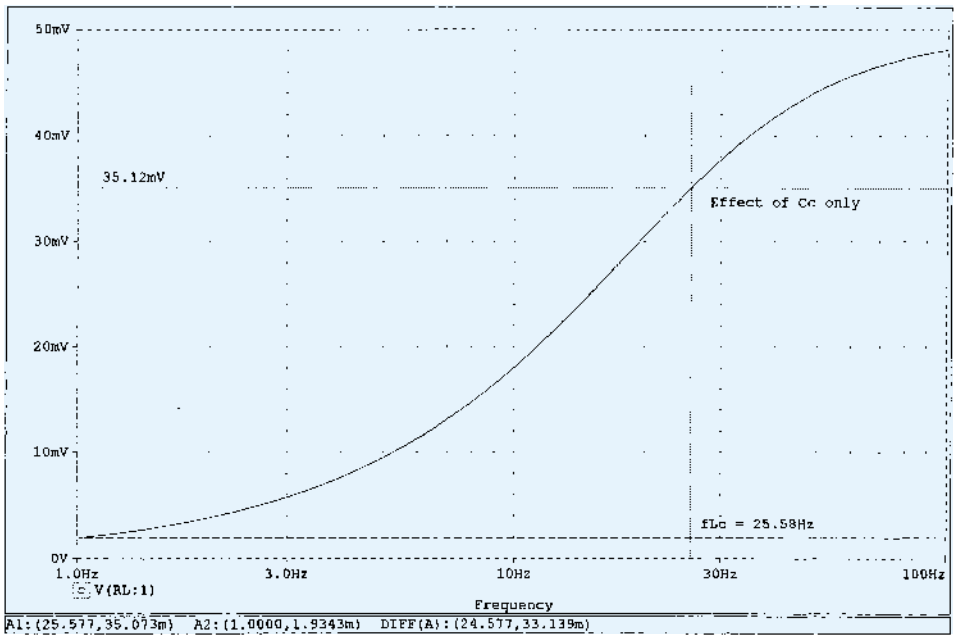


Figure 11.27 Low-frequency response due to C_C.

C_E

$$R_s' = R_s \| R_1 \| R_2 = 1\ \text{k}\Omega \| 40\ \text{k}\Omega \| 10\ \text{k}\Omega \cong 0.889\ \text{k}\Omega$$

$$R_e = R_E \left\| \left(\frac{R'_s}{\beta} + r_e \right) = 2\ \text{k}\Omega \right\| \left(\frac{0.889\ \text{k}\Omega}{100} + 15.76\ \Omega \right)$$

$$= 2\ \text{k}\Omega \| (8.89\ \Omega + 15.76\ \Omega) = 2\ \text{k}\Omega \| 24.65\ \Omega \cong 24.35\ \Omega$$

$$f_{L_E} = \frac{1}{2\pi R_e C_E} = \frac{1}{(6.28)(24.35\ \Omega)(20\ \mu\text{F})} = \frac{10^6}{3058.36} \cong \mathbf{327\ Hz}$$

The effect of C_E can be examined using PSpice Windows by setting both C_S and C_C to 1 Farad. In addition, since the frequency range is greater, the start frequency has to be changed to 10 Hz and the final frequency to 1 kHz. The result is the plot of Fig. 11.28, with a cutoff frequency of 321.17 Hz, providing a close match with the calculated value of 327 Hz.

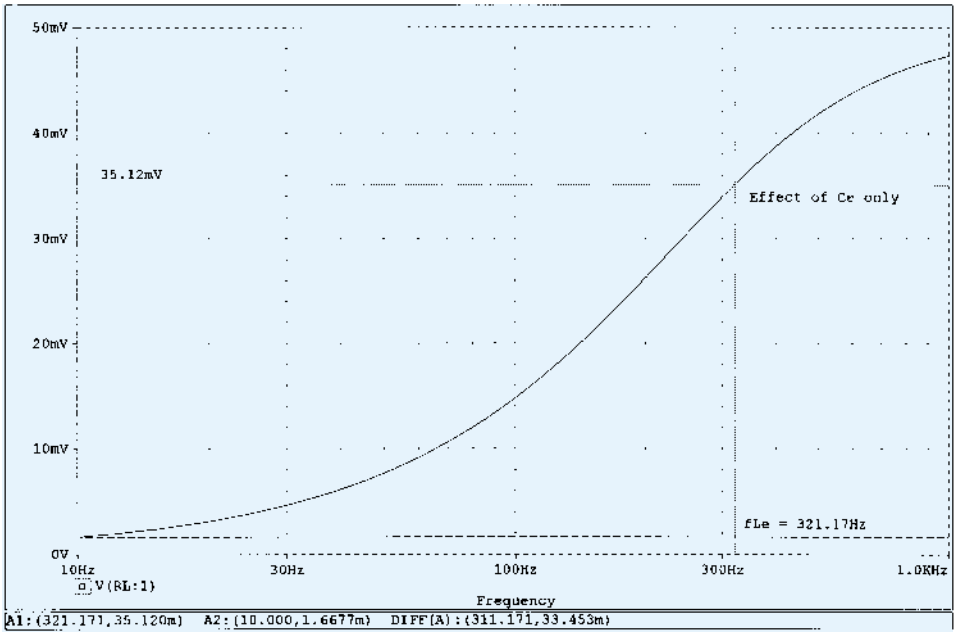


Figure 11.28 Low-frequency response due to C_E.

The fact that f_{L_E} is significantly higher than f_{L_S} or f_{L_C} suggests that it will be the predominant factor in determining the low-frequency response for the complete system. To test the accuracy of our hypothesis, the network is simulated with all the initial values of capacitance level to obtain the results of Fig. 11.29. Note the strong similarity with the waveform of Fig. 11.28, with the only visible difference being the higher gain at lower frequencies on Fig. 11.28. Without question, the plot supports the fact that the highest of the low cutoff frequencies will have the most impact on the low cutoff frequency for the system.

(b) It was mentioned earlier that dB plots are usually normalized by dividing the voltage gain A_v by the magnitude of the midband gain. For Fig. 11.16, the magnitude of the midband gain is 51.21, and naturally the ratio $|A_v/A_{v_{\text{mid}}}|$ will be 1 in the midband region. The result is a 0-dB asymptote in the midband region as shown in Fig. 11.30. Defining f_{L_E} as our lower cutoff frequency f_1, an asymptote at -6 dB/octave can be drawn as shown in Fig. 11.30 to form the Bode plot and our

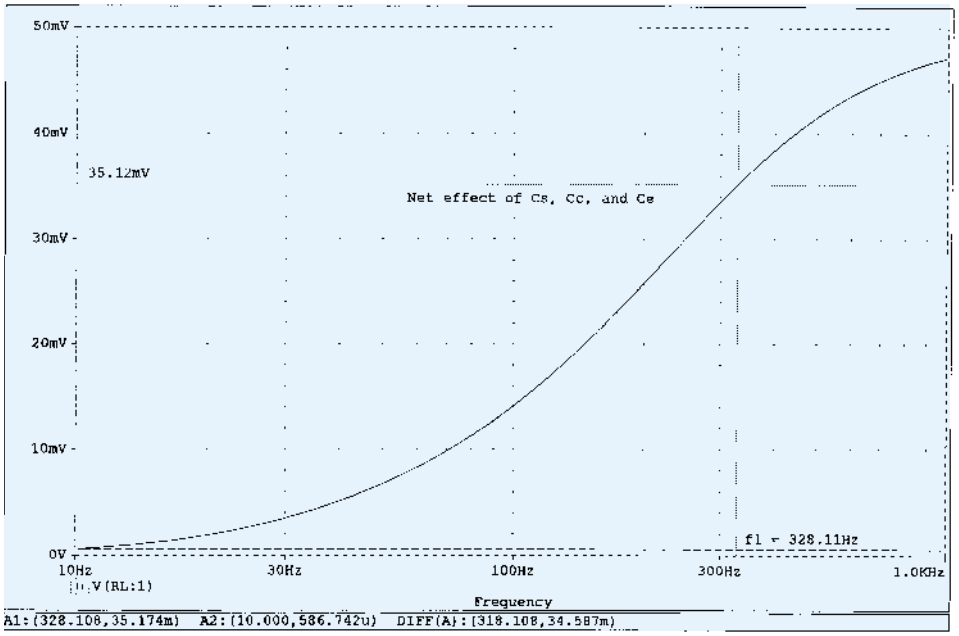


Figure 11.29 Low-frequency response due to C_S, C_E, and C_C.

envelope for the actual response. At f_1, the actual curve is -3 dB down from the midband level as defined by the $0.707A_{V_{mid}}$ level, permitting a sketch of the actual frequency response curve as shown in Fig. 11.30. A -6-dB/octave asymptote was drawn at each frequency defined in the analysis above to demonstrate clearly that it is f_{L_E} for this network that will determine the -3-dB point. It is not until about -24 dB that f_{L_C} begins to affect the shape of the envelope. The magnitude plot shows that the slope of the resultant asymptote is the sum of the asymptotes having the same sloping direction in the same frequency interval. Note in Fig. 11.30 that the slope has dropped to -12 dB/octave for frequencies less

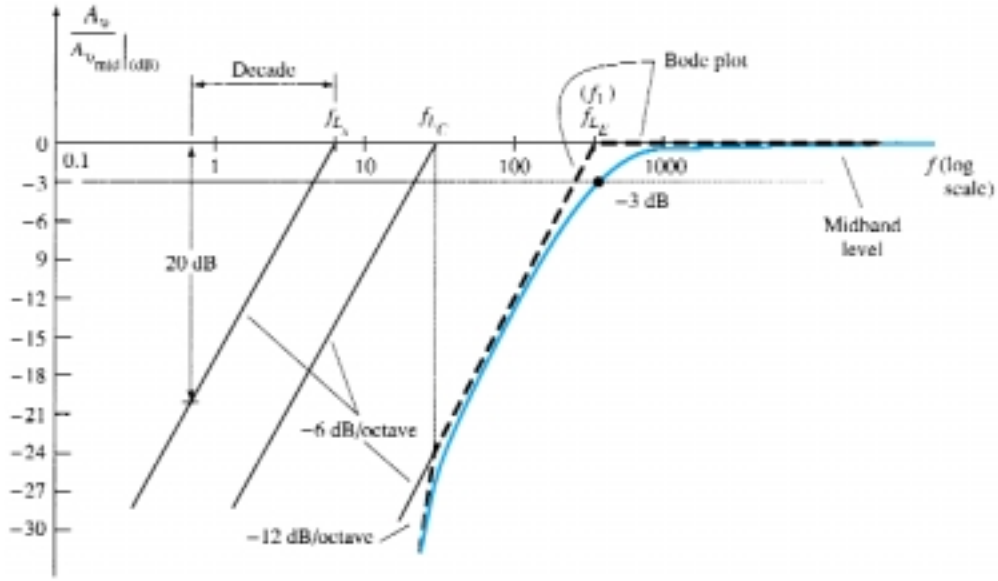


Figure 11.30 Low-frequency plot for the network of Example 11.9.

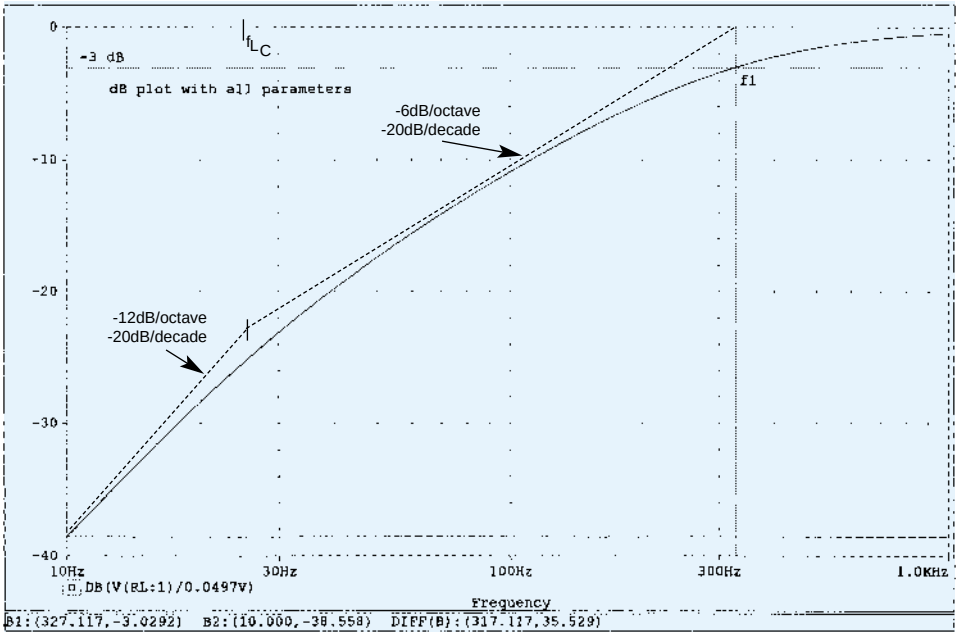


Figure 11.31 dB plot of the low-frequency response of the BJT amplifier of Fig. 11.25.

than f_{L_C} and could drop to -18 dB/octave if the three defined cutoff frequencies of Fig. 11.30 were closer together.

Using **PROBE,** a plot of $20 \log_{10}|A_v/A_{v_{mid}}| = A_v/A_{v_{mid}}|_{dB}$ can be obtained by recalling that if $V_s = 1$ mV, the magnitude of $|A_v/A_{v_{mid}}|$ is the same as $|V_o/A_{v_{mid}}|$ since V_o will have the same numerical value as A_v. The required **Trace Expression,** which is entered on the bottom of the **Add Traces** dialog box, appears on the horizontal axis of Fig. 11.31. The plot clearly reveals the change in slope of the asymptote at f_{L_C} and how the actual curve follows the envelope created by the Bode plot. In addition, note the 3-dB drop at f_1.

Keep in mind as we proceed to the next section that the analysis of this section is not limited to the network of Fig. 11.16. For any transistor configuration it is simply necessary to isolate each *R-C* combination formed by a capacitive element and determine the break frequencies. The resulting frequencies will then determine whether there is a strong interaction between capacitive elements in determining the overall response and which element will have the greatest impact on establishing the lower cutoff frequency. In fact, the analysis of the next section will parallel this section as we determine the low cutoff frequencies for the FET amplifier.

11.7 LOW-FREQUENCY RESPONSE — FET AMPLIFIER

The analysis of the FET amplifier in the low-frequency region will be quite similar to that of the BJT amplifier of Section 11.6. There are again three capacitors of primary concern as appearing in the network of Fig. 11.32: C_G, C_C, and C_S. Although Fig. 11.32 will be used to establish the fundamental equations, the procedure and conclusions can be applied to most FET configurations.

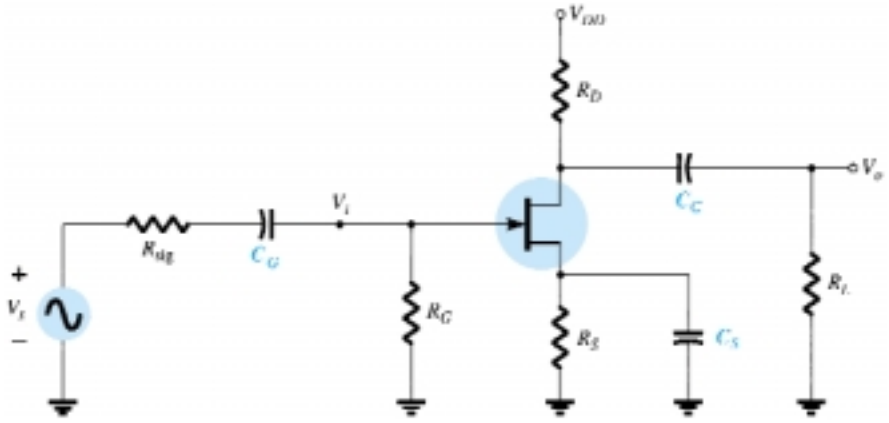


Figure 11.32 Capacitive elements that affect the low-frequency response of a JFET amplifier.

C_G

For the coupling capacitor between the source and the active device, the ac equivalent network will appear as shown in Fig. 11.33. The cutoff frequency determined by C_G will then be

$$f_{L_G} = \frac{1}{2\pi(R_{\text{sig}} + R_i)C_G} \tag{11.34}$$

which is an exact match of Eq. (11.26). For the network of Fig. 11.32,

$$R_i = R_G \tag{11.35}$$

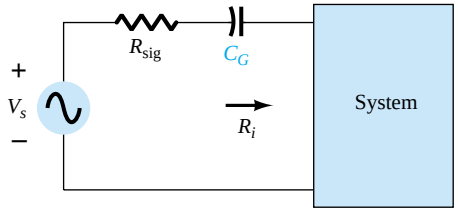


Figure 11.33 Determining the effect of C_G on the low-frequency response.

Typically, $R_G \gg R_{\text{sig}}$, and the lower cutoff frequency will be determined primarily by R_G and C_G. The fact that R_G is so large permits a relatively low level of C_G while maintaining a low cutoff frequency level for f_{L_G}.

C_C

For the coupling capacitor between the active device and the load the network of Fig. 11.34 will result, which is also an exact match of Fig. 11.19. The resulting cutoff frequency is

$$f_{L_C} = \frac{1}{2\pi(R_o + R_L)C_C} \tag{11.36}$$

For the network of Fig. 11.32,

$$R_o = R_D \| r_d \tag{11.37}$$

f

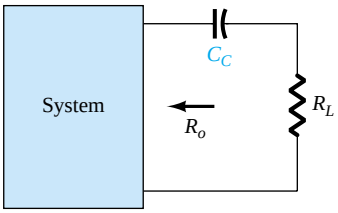


Figure 11.34 Determining the effect of C_C on the low-frequency response.

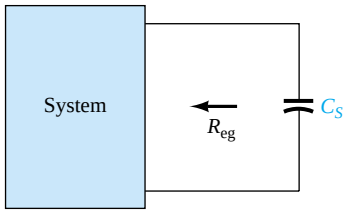


Figure 11.35 Determining the effect of C_S on the low-frequency response.

C_S

For the source capacitor C_S, the resistance level of importance is defined by Fig. 11.35. The cutoff frequency will be defined by

$$f_{L_S} = \frac{1}{2\pi R_{\text{eq}} C_S} \tag{11.38}$$

For Fig. 11.32, the resulting value of R_{eq}:

$$R_{\text{eq}} = \frac{R_S}{1 + R_S(1 + g_m r_d)/(r_d + R_D \| R_L)} \tag{11.39}$$

which for $r_d \cong \infty\ \Omega$ becomes

$$R_{\text{eq}} = R_S \| \frac{1}{g_m} \tag{11.40}$$

EXAMPLE 11.10

(a) Determine the lower cutoff frequency for the network of Fig. 11.32 using the following parameters:

$C_G = 0.01\ \mu\text{F}$, $C_C = 0.5\ \mu\text{F}$, $C_S = 2\ \mu\text{F}$

$R_{\text{sig}} = 10\ \text{k}\Omega$, $R_G = 1\ \text{M}\Omega$, $R_D = 4.7\ \text{k}\Omega$, $R_S = 1\ \text{k}\Omega$, $R_L = 2.2\ \text{k}\Omega$

$I_{DSS} = 8\text{mA}$, $V_P = -4\ \text{V}$ $r_d = \infty\ \Omega$, $\text{V}_{\text{DD}} = 20\ \text{V}$

(b) Sketch the frequency response using a Bode plot.

Solution

(a) DC Analysis: Plotting the transfer curve of $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ and superimposing the curve defined by $V_{GS} = -I_D R_S$ will result in an intersection at $V_{GS_Q} = -2$ V and $I_{D_Q} = 2$ mA. In addition,

$$g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(8\ \text{mA})}{4\ \text{V}} = 4\ \text{mS}$$

$$g_m = g_{m0}\left(1 - \frac{V_{GS_Q}}{V_P}\right) = 4\ \text{mS}\left(1 - \frac{-2\ \text{V}}{-4\ \text{V}}\right) = 2\ \text{mS}$$

C_G

$$\text{Eq. (11.34):}\quad f_{L_G} = \frac{1}{2\pi\,(10\ \text{k}\Omega + 1\ \text{M}\Omega)(0.01\ \mu\text{F})} \cong \mathbf{15.8\ Hz}$$

C_C

$$\text{Eq. (11.36):} \quad f_{L_C} = \frac{1}{2\pi\,(4.7\ \text{k}\Omega + 2.2\ \text{k}\Omega)(0.5\ \mu\text{F})} \cong \mathbf{46.13\ Hz}$$

C_S

$$R_{\text{eq}} = R_S \| \frac{1}{g_m} = 1\ \text{k}\Omega \| \frac{1}{2\ \text{mS}} = 1\ \text{k}\Omega \| 0.5\ \text{k}\Omega = 333.33\ \Omega$$

$$\text{Eq. (11.38):} \quad f_{L_S} = \frac{1}{2\pi\,(333.33\ \Omega)(2\ \mu\text{F})} = \mathbf{238.73\ Hz}$$

Since f_{L_S} is the largest of the three cutoff frequencies, it defines the low cutoff frequency for the network of Fig. 11.32.

(b) The midband gain of the system is determined by

$$A_{v_{\text{mid}}} = \frac{V_o}{V_i} = -g_m(R_D \| R_L) = -(2\ \text{mS})(4.7\ \text{k}\Omega \| 2.2\ \text{k}\Omega)$$

$$= -(2\ \text{mS})(1.499\ \text{k}\Omega)$$

$$\cong \mathbf{-3}$$

Using the midband gain to normalize the response for the network of Fig. 11.32 will result in the frequency plot of Fig. 11.36.

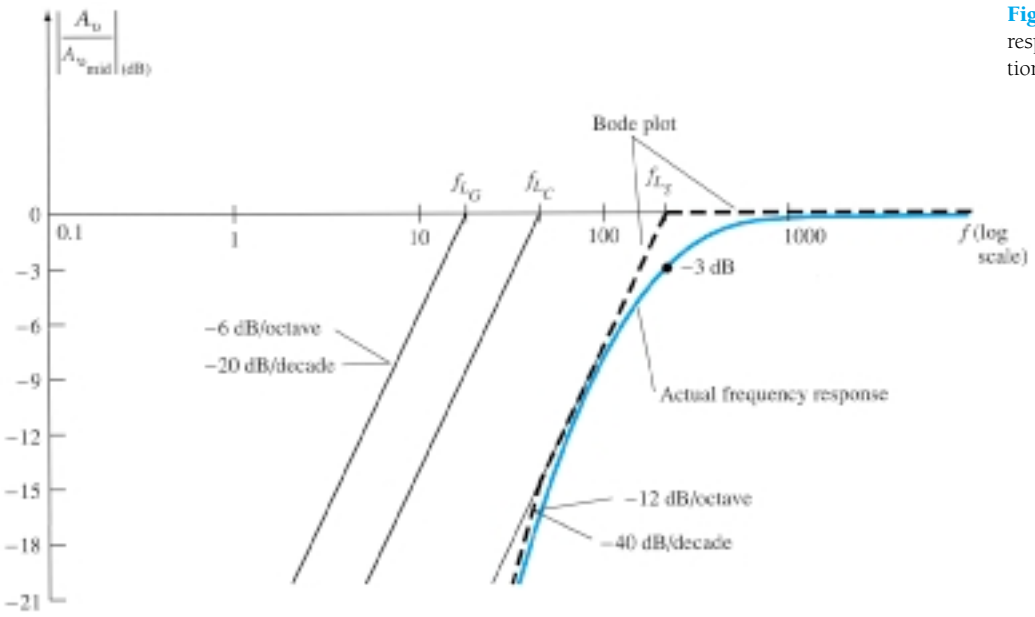


Figure 11.36 Low-frequency response for the JFET configuration of Example 11.10.

Using PSpice Windows, the network will appear as shown in Fig. 11.37, with the JFET parameters **Beta** set at 0.5mA/V^2 and **Vto** at -4 V (all others set to zero) and the frequency of interest at a midband value of 10 kHz. The resulting dc levels confirm that V_{GS} is -2 V and place V_D at 10.60 V, which should be right in the middle of the linear active region since $V_{GS} = 1/2(V_D = -4\ \text{V})$ and $V_{DS} = 1/2(V_{DD} = 20\ \text{V})$. The 0-V levels clearly reveal that the capacitors have isolated the transistor for the dc biasing. The ac response results in an ac level of 2.993 mV across the load for a gain of 2.993, which is essentially equal to the calculated gain of 3.

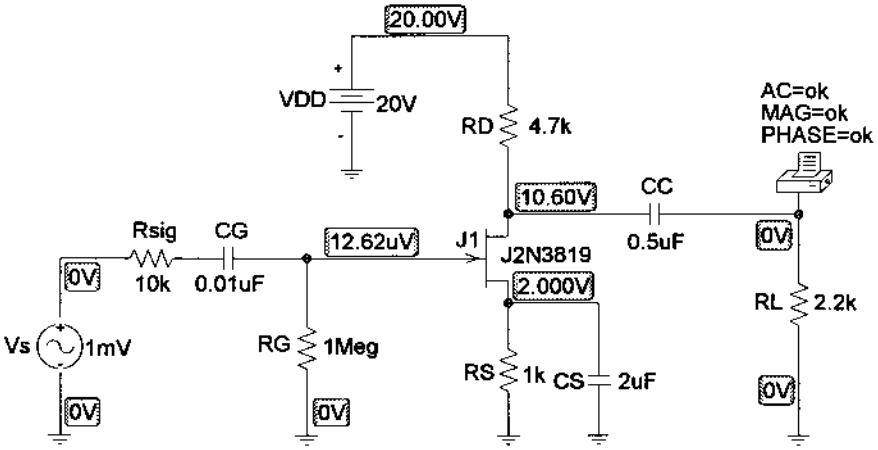


Figure 11.37 Schematic network for Example 11.10.

Returning to **Analysis** and choosing **Automatically run Probe after simulation** followed by **Setup-AC Sweep-Decade-Pts/Decade** = 1000, **Start Freq.:** 10Hz, and **End Freq.:** 10 kHz will setup **Simulation-Trace-Add-Trace Expression: DB (V(RL:1)/2.993mV)-OK,** which will result in the plot of Fig. 11.38, with a low cut-off frequency of 227.5 Hz primarily determined by the source capacitance.

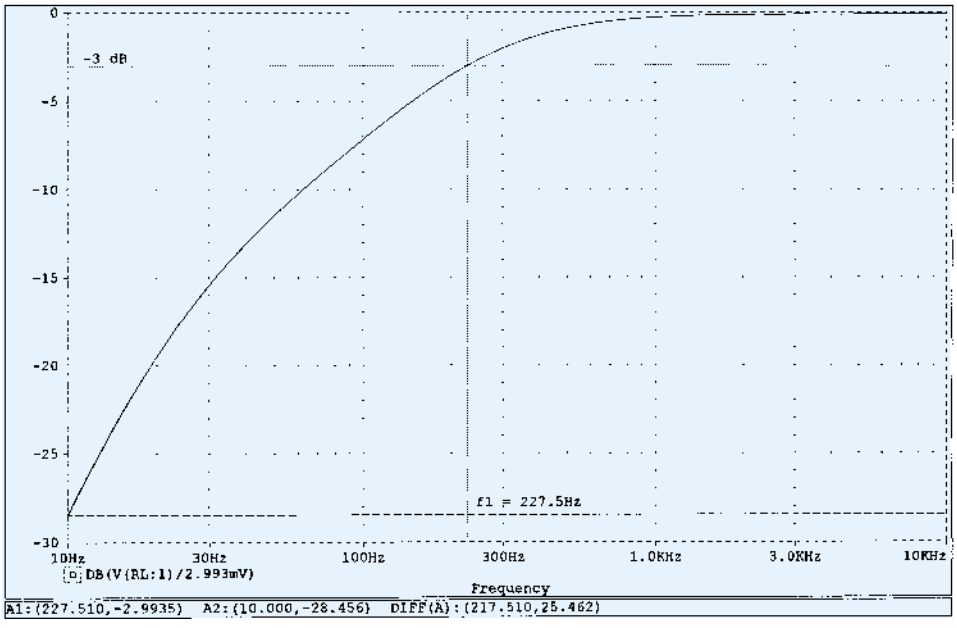


Figure 11.38 dB response for the low-frequency region in the network of Example 11.10.

11.8 MILLER EFFECT CAPACITANCE

In the high-frequency region, the capacitive elements of importance are the inter-electrode (between terminals) capacitances internal to the active device and the wiring capacitance between leads of the network. The large capacitors of the network that controlled the low-frequency response have all been replaced by their short-circuit equivalent due to their very low reactance levels.

For *inverting* amplifiers (phase shift of 180° between input and output resulting in a negative value for A_v), the input and output capacitance is increased by a capacitance level sensitive to the interelectrode capacitance between the input and output terminals of the device and the gain of the amplifier. In Fig. 11.39, this "feedback" capacitance is defined by C_f.

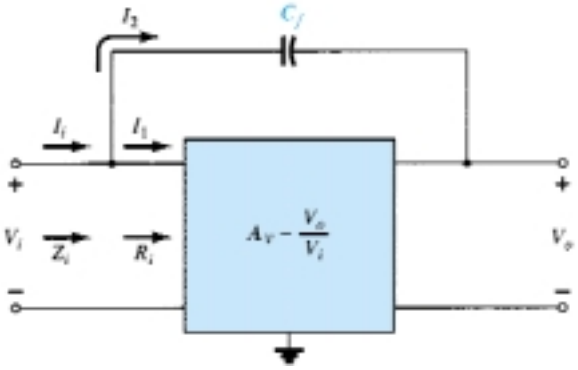


Figure 11.39 Network employed in the derivation of an equation for the Miller input capacitance.

Applying Kirchhoff's current law gives

$$I_i = I_1 + I_2$$

Using Ohm's law yields

$$I_i = \frac{V_i}{Z_i}, \qquad I_1 = \frac{V_i}{R_i}$$

and

$$I_2 = \frac{V_i - V_o}{X_{C_f}} = \frac{V_i - A_v V_i}{X_{C_f}} = \frac{(1 - A_v)V_i}{X_{C_f}}$$

Substituting, we obtain

$$\frac{V_i}{Z_i} = \frac{V_i}{R_i} + \frac{(1 - A_v)V_i}{X_{C_f}}$$

and

$$\frac{1}{Z_i} = \frac{1}{R_i} + \frac{1}{X_{C_f}/(1 - A_v)}$$

but

$$\frac{X_{C_f}}{1 - A_v} = \underbrace{\frac{1}{\omega\,(1 - A_v)C_f}}_{C_M} = X_{CM}$$

and

$$\frac{1}{Z_i} = \frac{1}{R_i} + \frac{1}{X_{C_M}}$$

establishing the equivalent network of Fig. 11.40. The result is an equivalent input impedance to the amplifier of Fig. 11.39 that includes the same R_i that we have dealt with in previous chapters, with the addition of a feedback capacitor magnified by the

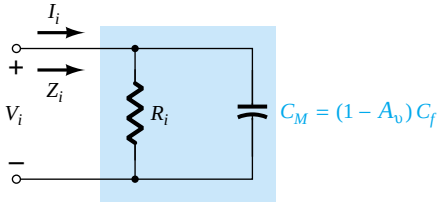


Figure 11.40 Demonstrating the impact of the Miller effect capacitance.

gain of the amplifier. Any interelectrode capacitance at the input terminals to the amplifier will simply be added in parallel with the elements of Fig. 11.40.

In general, therefore, the Miller effect input capacitance is defined by

$$C_{M_i} = (1 - A_v)C_f \tag{11.41}$$

This shows us that:

For any inverting amplifier, the input capacitance will be increased by a Miller effect capacitance sensitive to the gain of the amplifier and the interelectrode capacitance connected between the input and output terminals of the active device.

The dilemma of an equation such as Eq. (11.41) is that at high frequencies the gain A_v will be a function of the level of C_{M_i}. However, since the maximum gain is the midband value, using the midband value will result in the highest level of C_{M_i} and the worst-case scenario. In general, therefore, the midband value is typically employed for A_v in Eq. (11.41).

The reason for the constraint that the amplifier be of the inverting variety is now more apparent when one examines Eq. (11.41). A positive value for A_v would result in a negative capacitance (for $A_v > 1$).

The Miller effect will also increase the level of output capacitance, which must also be considered when the high-frequency cutoff is determined. In Fig. 11.41, the parameters of importance to determine the output Miller effect are in place. Applying Kirchhoff's current law will result in

$$I_o = I_1 + I_2$$

with $$I_1 = \frac{V_o}{R_o} \quad \text{and} \quad I_2 = \frac{V_o - V_i}{X_{C_f}}$$

The resistance R_o is usually sufficiently large to permit ignoring the first term of the equation compared to the second term and assuming that

$$I_o \cong \frac{V_o - V_i}{X_{C_f}}$$

Substituting $V_i = V_o/A_v$ from $A_v = V_o/V_i$ will result in

$$I_o = \frac{V_o - V_o/A_v}{X_{C_f}} = \frac{V_o(1 - 1/A_v)}{X_{C_f}}$$

and

$$\frac{I_o}{V_o} = \frac{1 - 1/A_v}{X_{C_f}}$$

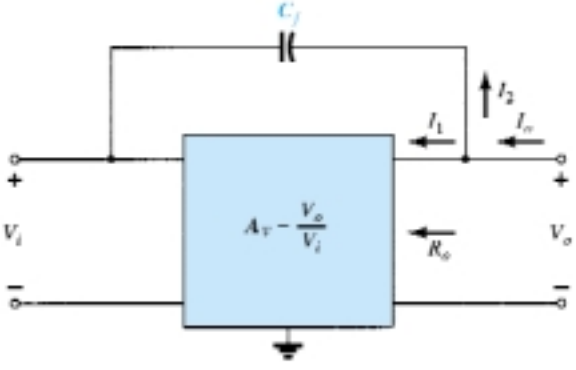


Figure 11.41 Network employed in the derivation of an equation for the Miller output capacitance.

or
$$\frac{V_o}{I_o} = \frac{X_{C_f}}{1 - 1/A_v} = \frac{1}{\omega C_f (1 - 1/A_v)} = \frac{1}{\omega C_{M_o}}$$

resulting in the following equation for the Miller output capacitance:

$$C_{M_o} = \left(1 - \frac{1}{A_v}\right) C_f \tag{11.42a}$$

For the usual situation where $A_v \gg 1$, Eq. (11.42a) reduces to

$$C_{M_o} \cong C_f \quad |A_v| \gg 1 \tag{11.42b}$$

Examples in the use of Eq. (11.42) will appear in the next two sections as we investigate the high-frequency responses of BJT and FET amplifiers.

11.9 HIGH-FREQUENCY RESPONSE — BJT AMPLIFIER

At the high-frequency end, there are two factors that will define the −3-dB point: the network capacitance (parasitic and introduced) and the frequency dependence of $h_{fe}(\beta)$.

Network Parameters

In the high-frequency region, the *RC* network of concern has the configuration appearing in Fig. 11.42. At increasing frequencies, the reactance X_C will decrease in magnitude, resulting in a shorting effect across the output and a decrease in gain. The derivation leading to the corner frequency for this *RC* configuration follows along similar lines to that encountered for the low-frequency region. The most significant difference is in the general form of A_v appearing below:

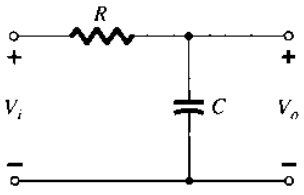


Figure 11.42 *R-C* combination that will define a high cutoff frequency.

$$A_v = \frac{1}{1 + j(f/f_2)} \tag{11.43}$$

which results in a magnitude plot such as shown in Fig. 11.43 that drops off at 6 dB/octave with increasing frequency. Note that f_2 is in the denominator of the frequency ratio rather than the numerator as occurred for f_1 in Eq. (11.21).

In Fig. 11.44, the various parasitic capacitances (C_{be}, C_{bc}, C_{ce}) of the transistor

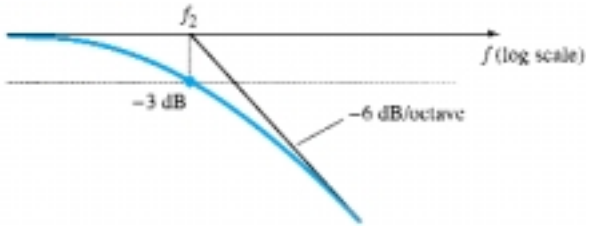


Figure 11.43 Asymptotic plot as defined by Eq. (11.43).

have been included with the wiring capacitances (C_{W_i}, C_{W_o}) introduced during construction. The high-frequency equivalent model for the network of Fig. 11.44 appears in Fig. 11.45. Note the absence of the capacitors C_s, C_C, and C_E, which are all assumed to be in the short-circuit state at these frequencies. The capacitance C_i includes

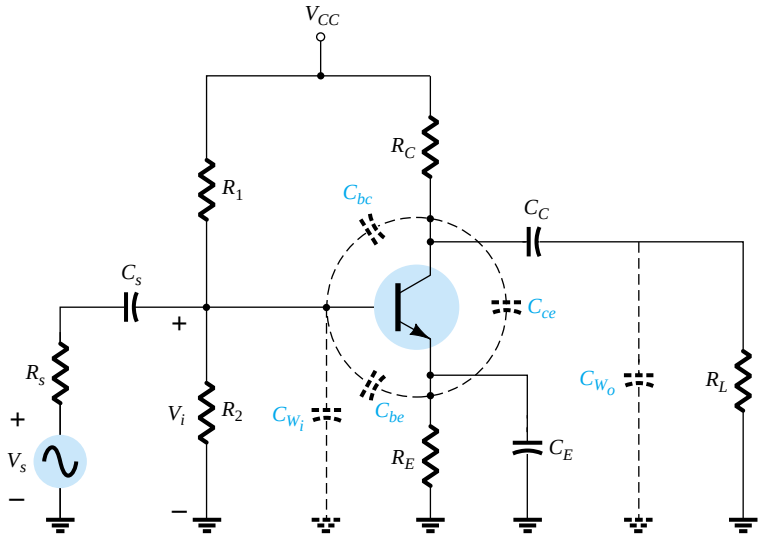


Figure 11.44 Network of Fig. 11.16 with the capacitors that affect the high-frequency response.

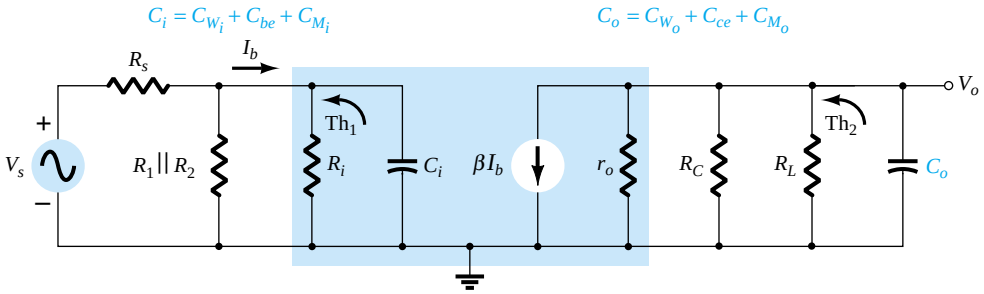


Figure 11.45 High-frequency ac equivalent model for the network of Fig. 11.44.

the input wiring capacitance C_{W_i}, the transition capacitance C_{be}, and the Miller capacitance C_{M_i}. The capacitance C_o includes the output wiring capacitance C_{W_o}, the parasitic capacitance C_{ce}, and the output Miller capacitance C_{M_o}. In general, the capacitance C_{be} is the largest of the parasitic capacitances, with C_{ce} the smallest. In fact, most specification sheets simply provide the levels of C_{be} and C_{bc} and do not include C_{ce} unless it will affect the response of a particular type of transistor in a specific area of application.

Determining the Thévenin equivalent circuit for the input and output networks of Fig. 11.45 will result in the configurations of Fig. 11.46. For the input network, the −3-dB frequency is defined by

$$f_{H_i} = \frac{1}{2\pi R_{\mathrm{Th}_1} C_i} \tag{11.44}$$

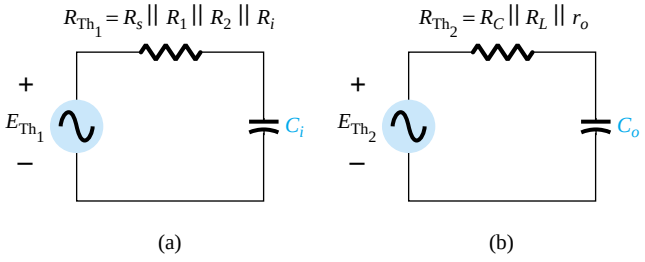


Figure 11.46 Thévenin circuits for the input and output networks of the network of Fig. 11.45.

with $$R_{\text{Th}_1} = R_s \| R_1 \| R_2 \| R_i \tag{11.45}$$

and $$C_i = C_{W_i} + C_{be} + C_{M_i} = C_{W_i} + C_{be} + (1 - A_v)C_{bc} \tag{11.46}$$

At very high frequencies, the effect of C_i is to reduce the total impedance of the parallel combination of R_1, R_2, R_i, and C_i in Fig. 11.45. The result is a reduced level of voltage across C_i, a reduction in I_b, and a gain for the system.

For the output network,

$$f_{H_o} = \frac{1}{2\pi R_{\text{Th}_2} C_o} \tag{11.47}$$

with $$R_{\text{Th}_2} = R_C \| R_L \| r_o \tag{11.48}$$

and $$C_o = C_{W_o} + C_{ce} + C_{M_o} \tag{11.49}$$

At very high frequencies, the capacitive reactance of C_o will decrease and consequently reduce the total impedance of the output parallel branches of Fig. 11.45. The net result is that V_o will also decline toward zero as the reactance X_C becomes smaller. The frequencies f_{H_i} and f_{H_o} will each define a -6-dB/octave asymptote such as depicted in Fig. 11.43. If the parasitic capacitors were the only elements to determine the high cutoff frequency, the lowest frequency would be the determining factor. However, the decrease in h_{fe} (or β) with frequency must also be considered as to whether its break frequency is lower than f_{H_i} or f_{H_o}.

h_{fe} (or β) Variation

The variation of h_{fe} (or β) with frequency will approach, with some degree of accuracy, the following relationship:

$$h_{fe} = \frac{h_{fe_{\text{mid}}}}{1 + j(f/f_\beta)} \tag{11.50}$$

The use of h_{fe} rather than β in some of this descriptive material is due primarily to the fact that manufacturers typically use the hybrid parameters when covering this issue in their specification sheets, and so on.

The only undefined quantity, f_β, is determined by a set of parameters employed in the *hybrid* π or *Giacoletto* model frequently applied to best represent the transistor in the high-frequency region. It appears in Fig. 11.47. The various parameters warrant a moment of explanation. The resistance $r_{bb'}$ includes the base contact, base bulk, and base spreading resistance. The first is due to the actual connection to the base. The second includes the resistance from the external terminal to the active region of

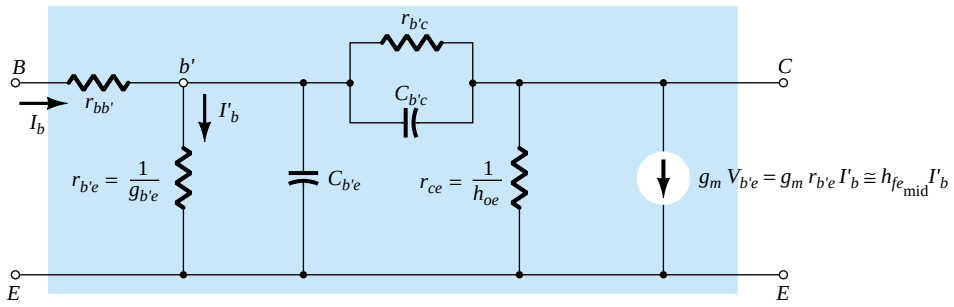


Figure 11.47 Giacoletto (or hybrid π) high-frequency transistor small-signal ac equivalent circuit.

the transistors, while the last is the actual resistance within the active base region. The resistances $r_{b'e}$, r_{ce}, and $r_{b'c}$ are the resistances between the indicated terminals when the device is in the active region. The same is true for the capacitances $C_{b'c}$ and $C_{b'e}$, although the former is a transition capacitance while the latter is a diffusion capacitance. A more detailed explanation of the frequency dependence of each can be found in a number of readily available texts.

In terms of these parameters,

$$f_\beta \text{ (sometimes appearing as } f_{h_{fe}}) = \frac{g_{b'e}}{2\pi(C_{b'e} + C_{b'c})} \tag{11.51}$$

or since the hybrid parameter h_{fe} is related to $g_{b'e}$ through $g_m = h_{fe_{\text{mid}}} g_{b'e}$,

$$f_\beta = \frac{1}{h_{fe_{\text{mid}}}} \frac{g_m}{2\pi(C_{b'e} + C_{b'c})} \tag{11.52}$$

Taking it a step further,

$$g_m = h_{fe_{\text{mid}}} g_{b'e} = h_{fe_{\text{mid}}} \frac{1}{r_{b'e}} \cong \frac{h_{fe_{\text{mid}}}}{h_{ie}} = \frac{\beta_{\text{mid}}}{\beta_{\text{mid}} r_e} = \frac{1}{r_e}$$

and using the approximations

$$C_{b'e} \cong C_{be} \qquad \text{and} \qquad C_{b'c} \cong C_{bc}$$

will result in the following form for Eq. (11.50):

$$f_\beta \cong \frac{1}{2\pi\beta_{\text{mid}} r_e (C_{be} + C_{bc})} \tag{11.53}$$

Equation (11.53) clearly reveals that since r_e is a function of the network design:

f_β is a function of the bias conditions.

The basic format of Eq. (11.50) is exactly the same as Eq. (11.43) if we extract the multiplying factor $h_{fe_{\text{mid}}}$, revealing that h_{fe} will drop off from its midband value with a 6-dB/octave slope as shown in Fig. 11.48. The same figure has a plot of h_{fb} (or α) versus frequency. Note the small change in h_{fb} for the chosen frequency range, revealing that the common-base configuration displays improved high-frequency characteristics over the common-emitter configuration. Recall also the absence of the Miller effect capacitance due to the noninverting characteristics of the common-base configuration. For this very reason, common-base high-frequency parameters rather than common-emitter parameters are often specified for a transistor—especially those designed specifically to operate in the high-frequency regions.

The following equation permits a direct conversion for determining f_β if f_α and α are specified.

$$f_\beta = f_\alpha(1 - \alpha) \tag{11.54}$$

A quantity called the *gain–bandwidth product* is defined for the transistor by the condition

$$\left| \frac{h_{fe_{\text{mid}}}}{1 + j(f/f_\beta)} \right| = 1$$

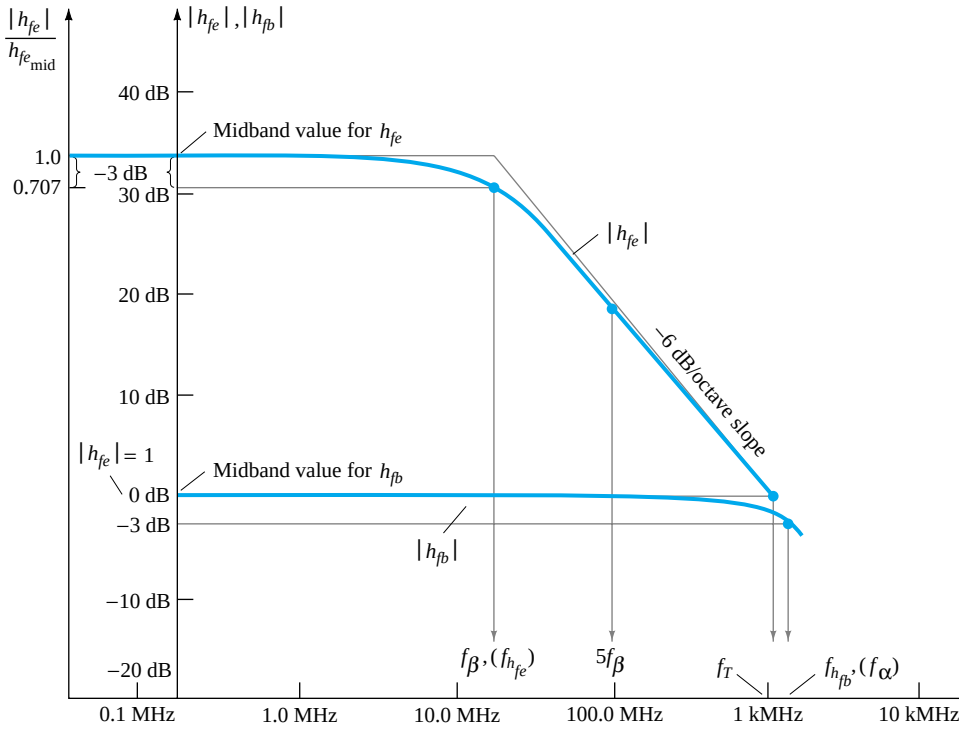


Figure 11.48 h_{fe} and h_{fb} versus frequency in the high-frequency region.

so that $$|h_{fe}|_{dB} = 20 \log_{10} \left| \frac{h_{fe_{mid}}}{1 + j(f/f_\beta)} \right| = 20 \log_{10} 1 = 0 \text{ dB}$$

The frequency at which $|h_{fe}|_{dB} = 0$ dB is clearly indicated by f_T in Fig. 11.48. The magnitude of h_{fe} at the defined condition point ($f_T \gg f_\beta$) is given by

$$\frac{h_{fe_{mid}}}{\sqrt{1 + (f_T/f_\beta)^2}} \cong \frac{h_{fe_{mid}}}{f_T/f_\beta} = 1$$

so that $$\boxed{f_T \cong h_{fe_{mid}} \cdot \overbrace{f_\beta}^{(\cong \text{BW})}} \quad \text{(gain–bandwidth product)} \qquad (11.55)$$

or $$\boxed{f_T \cong \beta_{mid} f_\beta} \qquad (11.56)$$

with $$\boxed{f_\beta = \frac{f_T}{\beta_{mid}}} \qquad (11.57)$$

Substituting Eq. (11.53) for f_β in Eq. (11.55) gives

$$f_T \cong \beta_{mid} \frac{1}{2\pi\beta_{mid} r_e (C_{be} + C_{bc})}$$

and $$\boxed{f_T \cong \frac{1}{2\pi r_e (C_{be} + C_{bc})}} \qquad (11.58)$$

EXAMPLE 11.11

For the network of Fig. 11.44 with the same parameters as in Example 11.9, that is,

$R_s = 1\ \text{k}\Omega$, $R_1 = 40\ \text{k}\Omega$, $R_2 = 10\ \text{k}\Omega$, $R_E = 2\ \text{k}\Omega$, $R_C = 4\ \text{k}\Omega$, $R_L = 2.2\ \text{k}\Omega$

$C_s = 10\ \mu\text{F}$, $C_C = 1\ \mu\text{F}$, $C_E = 20\ \mu\text{F}$

$\beta = 100$, $r_o = \infty\ \Omega$, $V_{CC} = 20\ \text{V}$

with the addition of

$$C_{be} = 36\ \text{pF},\ C_{bc} = 4\ \text{pF},\ C_{ce} = 1\ \text{pF},\ C_{W_i} = 6\ \text{pF},\ C_{W_o} = 8\ \text{pF}$$

(a) Determine f_{H_i} and f_{H_o}.
(b) Find f_β and f_T.
(c) Sketch the frequency response for the low- and high-frequency regions using the results of Example 11.9 and the results of parts (a) and (b).
(d) Obtain a **PROBE** response for the full frequency spectrum and compare with the results of part (c).

Solution

(a) From Example 11.9:

$$R_i = 1.32\ \text{k}\Omega, \qquad A_{v_{\text{mid}}}(\text{amplifier}) = -90$$

and

$$R_{\text{Th}_1} = R_s \| R_1 \| R_2 \| R_i = 1\ \text{k}\Omega \| 40\ \text{k}\Omega \| 10\ \text{k}\Omega \| 1.32\ \text{k}\Omega$$
$$\cong 0.531\ \text{k}\Omega$$

with

$$C_i = C_{W_i} + C_{be} + (1 - A_v)C_{be}$$
$$= 6\ \text{pF} + 36\ \text{pF} + [1 - (-90)]4\ \text{pF}$$
$$= 406\ \text{pF}$$

$$f_{H_i} = \frac{1}{2\pi R_{\text{Th}_1} C_i} = \frac{1}{2\pi(0.531\ \text{k}\Omega)(406\ \text{pF})}$$
$$= \mathbf{738.24\ kHz}$$

$$R_{\text{Th}_2} = R_C \| R_L = 4\ \text{k}\Omega \| 2.2\ \text{k}\Omega = 1.419\ \text{k}\Omega$$

$$C_o = C_{W_o} + C_{ce} + C_{M_o} = 8\ \text{pF} + 1\ \text{pF} + \left(1 - \frac{1}{-90}\right) 4\ \text{pF}$$
$$= 13.04\ \text{pF}$$

$$f_{H_o} = \frac{1}{2\pi R_{\text{Th}_2} C_o} = \frac{1}{2\pi(1.419\ \text{k}\Omega)(13.04\ \text{pF})}$$
$$= \mathbf{8.6\ MHz}$$

(b) Applying Eq. (11.53) gives

$$f_\beta = \frac{1}{2\pi\beta_{\text{mid}} r_e (C_{be} + C_{bc})}$$

$$= \frac{1}{2\pi(100)(15.76\ \Omega)(36\ \text{pF} + 4\ \text{pF})} = \frac{1}{2\pi(100)(15.76\ \Omega)(40\ \text{pF})}$$

$$= \mathbf{2.52\ MHz}$$

$$f_T = \beta_{\text{mid}}\, f_\beta = (100)(2.52\ \text{MHz})$$

$$= \mathbf{252\ MHz}$$

(c) See Fig. 11.49. Both f_β and f_{H_o} will lower the upper cutoff frequency below the level determined by f_{H_i}. f_β is closer to f_{H_i} and therefore will have a greater impact than f_{H_o}. In any event, the bandwidth will be less than that defined solely by f_{H_i}. In fact, for the parameters of this network the upper cutoff frequency will be relatively close to 600 kHz.

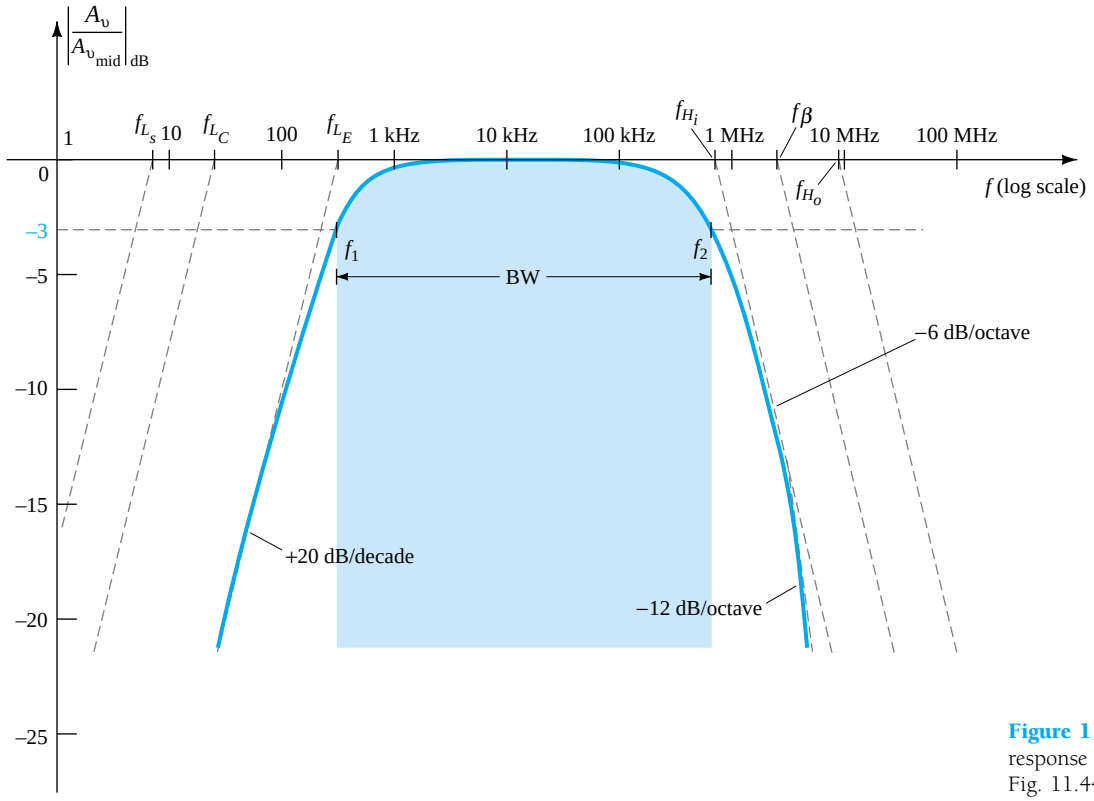


Figure 11.49 Full frequency response for the network of Fig. 11.44.

In general, therefore, the lowest of the upper-cutoff frequencies defines a maximum possible bandwidth for a system.

(d) In order to obtain a PSpice analysis for the full frequency range, the parasitic capacitances have to be added to the network as shown in Fig. 11.50.

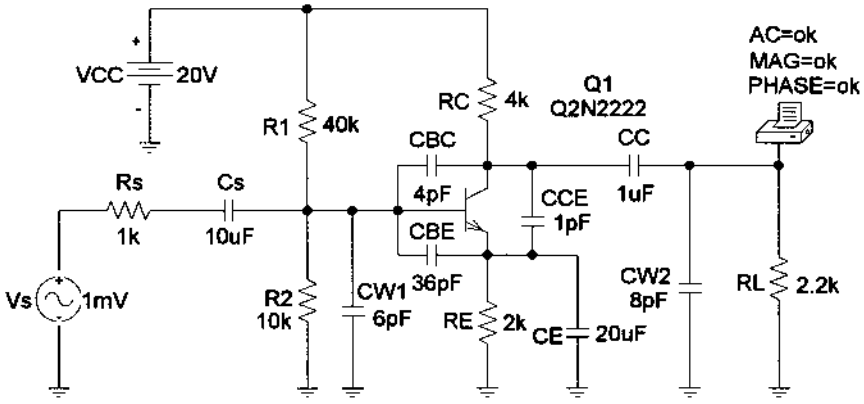


Figure 11.50 Network of Figure 11.25 with parasitic capacitances in place.

f

An **Analysis** will result in the plot of Fig. 11.51 using the **Trace Expression** appearing at the bottom of the plot. The vertical scale was changed from −60 to 0 dB to −30 to 0 dB to highlight the area of interest using the **Y-Axis Settings.** The low cutoff frequency of 324 Hz is as determined primarily by f_{L_E}, and the high cutoff frequency is near 667kHz. Even though f_{H_o} is more than a decade higher than f_{H_i}, it will have an impact on the high cutoff frequency. In total, however, the PSpice analysis has been a welcome verification of the hand-written approach.

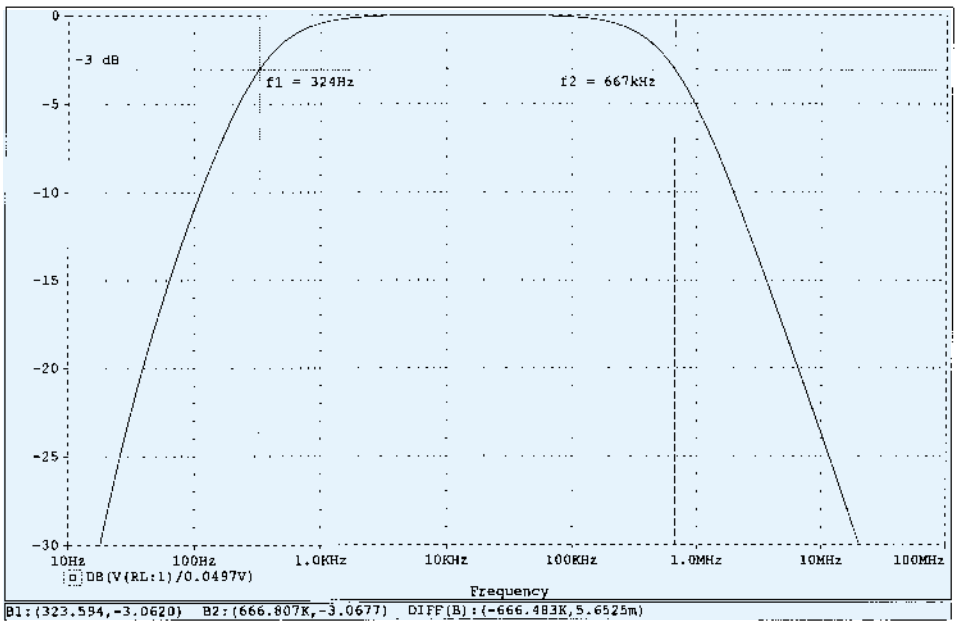


Figure 11.51 Full frequency response for the network of Fig. 11.50.

11.10 HIGH-FREQUENCY RESPONSE — FET AMPLIFIER

The analysis of the high-frequency response of the FET amplifier will proceed in a very similar manner to that encountered for the BJT amplifier. As shown in Fig. 11.52, there are interelectrode and wiring capacitances that will determine the high-frequency characteristics of the amplifier. The capacitors C_{gs} and C_{gd} typically vary from 1 to 10 pF, while the capacitance C_{ds} is usually quite a bit smaller, ranging from 0.1 to 1 pF.

Since the network of Fig. 11.52 is an inverting amplifier, a Miller effect capacitance will appear in the high-frequency ac equivalent network appearing in Fig. 11.53. At high frequencies, C_i will approach a short-circuit equivalent and V_{gs} will drop in value and reduce the overall gain. At frequencies where C_o approaches its short-circuit equivalent, the parallel output voltage V_o will drop in magnitude.

The cutoff frequencies defined by the input and output circuits can be obtained by first finding the Thévenin equivalent circuits for each section as shown in Fig. 11.54. For the input circuit,

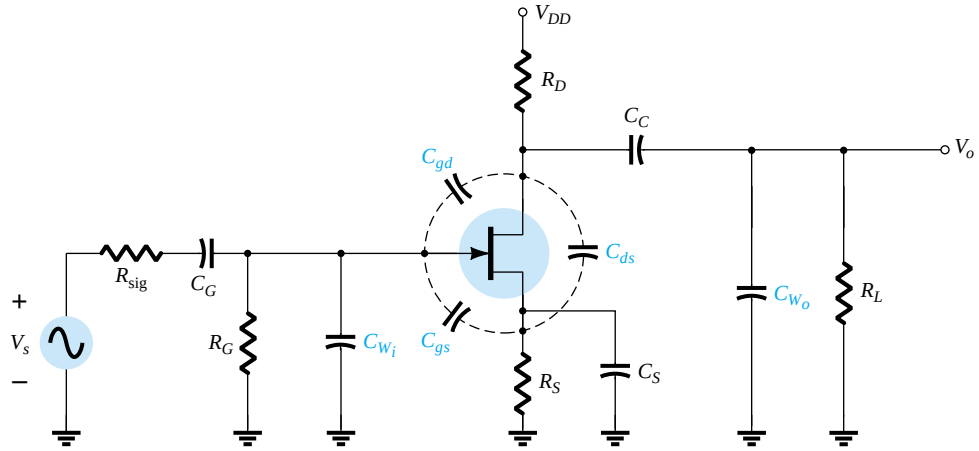


Figure 11.52 Capacitive elements that affect the high frequency response of a JFET amplifier.

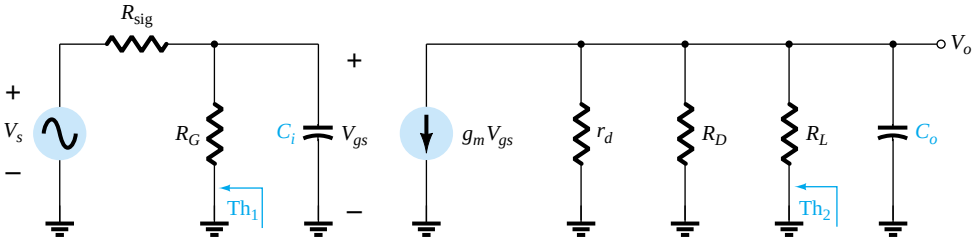


Figure 11.53 High-frequency ac equivalent circuit for Fig. 11.52.

$$f_{H_i} = \frac{1}{2\pi R_{\text{Th}_1} C_i} \tag{11.59}$$

and

$$R_{\text{Th}_1} = R_{\text{sig}} \| R_G \tag{11.60}$$

with

$$C_i = C_{W_i} + C_{gs} + C_{M_i} \tag{11.61}$$

and

$$C_{M_i} = (1 - A_v) C_{gd} \tag{11.62}$$

and for the output circuit,

$$f_{H_o} = \frac{1}{2\pi R_{\text{Th}_2} C_o} \tag{11.63}$$

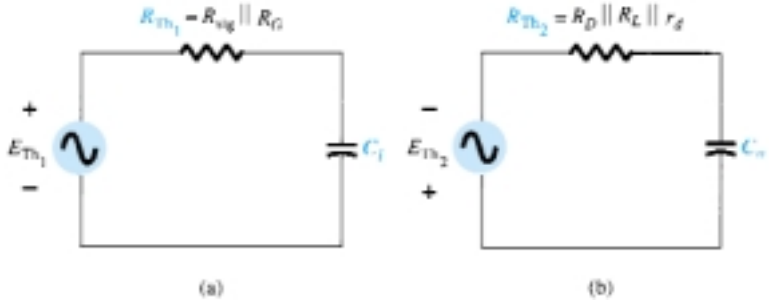


Figure 11.54 The Thévenin equivalent circuits for the (a) input circuit and (b) output circuit.

f

with $$R_{\text{Th}_2} = R_D \| R_L \| r_d \tag{11.64}$$

and $$C_o = C_{W_o} + C_{ds} + C_{M_o}$$

and $$C_{M_o} = \left(1 - \frac{1}{Av}\right) C_{gd} \tag{11.65}$$

EXAMPLE 11.12

(a) Determine the high cutoff frequencies for the network of Fig. 11.52 using the same parameters as Example 11.10:

$$C_G = 0.01\ \mu\text{F}, \quad C_C = 0.5\ \mu\text{F}, \quad C_S = 2\ \mu\text{F}$$

$$R_{\text{sig}} = 10\ \text{k}\Omega, \quad R_G = 1\ \text{M}\Omega, \quad R_D = 4.7\ \text{k}\Omega, \quad R_S = 1\ \text{k}\Omega, \quad R_L = 2.2\ \text{k}\Omega$$

$$I_{DSS} = 8\ \text{mA}, \quad V_P = -4\ \text{V}, \quad r_d = \infty\ \Omega, \quad V_{DD} = 20\ \text{V}$$

with the addition of

$$C_{gd} = 2\ \text{pF}, \quad C_{gs} = 4\ \text{pF}, \quad C_{ds} = 0.5\ \text{pF}, \quad C_{W_i} = 5\ \text{pF}, \quad C_{W_o} = 6\ \text{pF}$$

(b) Review a **PROBE** response for the full frequency range and note whether it supports the conclusions of Example 11.10 and the calculations above.

Solution

(a) $R_{\text{Th}_1} = R_{\text{sig}} \| R_G = 10\ \text{k}\Omega \| 1\ \text{M}\Omega = 9.9\ \text{k}\Omega$

From Example 11.10, $A_v = -3$.

$$\begin{aligned} C_i &= C_{W_i} + C_{gs} + (1 - A_v)C_{gd} \\ &= 5\ \text{pF} + 4\ \text{pF} + (1 + 3)2\ \text{pF} \\ &= 9\ \text{pF} + 8\ \text{pF} \\ &= 17\ \text{pF} \end{aligned}$$

$$f_{H_i} = \frac{1}{2\pi R_{Th_1} C_i}$$

$$= \frac{1}{2\pi(9.9\ \text{k}\Omega)(17\ \text{pF})} = \mathbf{945.67\ kHz}$$

$$\begin{aligned} R_{\text{Th}_2} &= R_D \| R_L \\ &= 4.7\ \text{k}\Omega \| 2.2\ \text{k}\Omega \\ &\cong 1.5\ \text{k}\Omega \end{aligned}$$

$$C_o = C_{W_o} + C_{ds} + C_{M_o} = 6\ \text{pF} + 0.5\ \text{pF} + \left(1 - \frac{1}{-3}\right)2\ \text{pF} = 9.17\ \text{pF}$$

$$f_{H_o} = \frac{1}{2\pi(1.5\ \text{k}\Omega)(9.17\ \text{pF})} = \mathbf{11.57\ MHz}$$

The results above clearly indicate that the input capacitance with its Miller effect capacitance will determine the upper cutoff frequency. This is typically the case due to the smaller value of C_{ds} and the resistance levels encountered in the output circuit.

(b) Using PSpice Windows, the schematic for the network will appear as shown in Fig. 11.55.

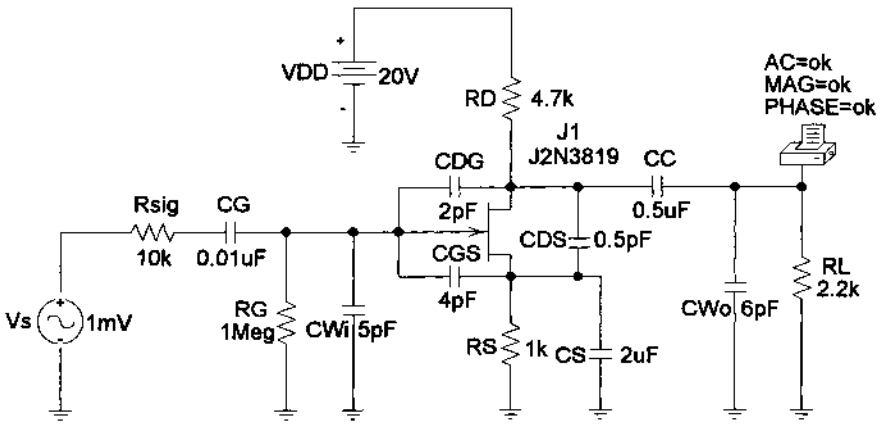


Figure 11.55 Network of Figure 11.52 with assigned values.

Under **Analysis,** the **AC Sweep** is set to **Decade** with **Pts/Decade** at 1000, **Start Freq.:** at 10 Hz, and **End Freq.:** at 10 MHz. Under the **Add Traces** dialog box, the **Trace Expression** is entered as **DB(V(RL:1)/2.993mV),** and the plot of Fig. 11.56 is obtained. Just for a moment, consider how much time it must have taken to obtain a plot such as in Fig. 11.56 without computer methods for a network as complicated as Fig. 11.55. Often, we forget how computer systems have helped us through some painstaking, lengthy, and boring series of calculations.

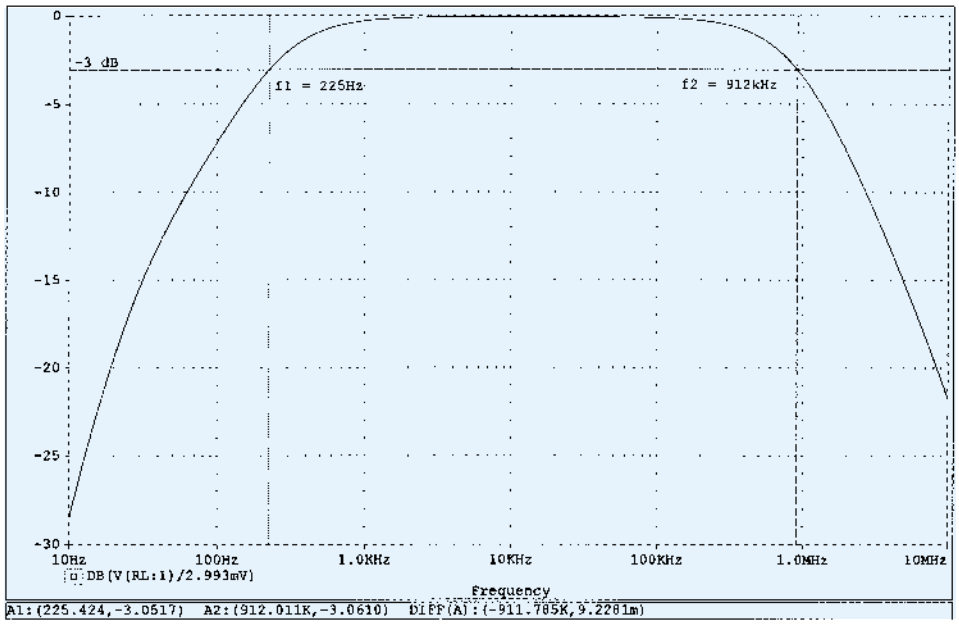


Figure 11.56 Frequency response for the network of Example 11.12.

Using the cursor, we find the lower and upper cutoff frequencies to be 225 Hz and 921 kHz, respectively, providing a nice match with the calculated values.

Even though the analysis of the past few sections has been limited to two configurations, the exposure to the general procedure for determining the cutoff frequencies should support the analysis of any other transistor configuration. Keep in mind that the Miller capacitance is limited to inverting amplifiers and that f_α is significantly greater than f_β if the common-base configuration is encountered. There is a great deal more literature on the analysis of single-stage amplifiers that goes beyond the coverage of this chapter. However, the content of this chapter should provide a firm foundation for any future analysis of frequency effects.

11.11 MULTISTAGE FREQUENCY EFFECTS

For a second transistor stage connected directly to the output of a first stage, there will be a significant change in the overall frequency response. In the high-frequency region, the output capacitance C_o must now include the wiring capacitance (C_{W_1}), parasitic capacitance (C_{be}), and Miller capacitance (C_{M_i}) of the following stage. Further, there will be additional low-frequency cutoff levels due to the second stage that will further reduce the overall gain of the system in this region. For each additional stage, the upper cutoff frequency will be determined primarily by that stage having the lowest cutoff frequency. The low-frequency cutoff is primarily determined by that stage having the highest low-frequency cutoff frequency. Obviously, therefore, one poorly designed stage can offset an otherwise well-designed cascaded system.

The effect of increasing the number of *identical* stages can be clearly demonstrated by considering the situations indicated in Fig. 11.57. In each case, the upper and lower cutoff frequencies of each of the cascaded stages are identical. For a single stage, the cutoff frequencies are f_1 and f_2 as indicated. For two identical stages in cascade, the drop-off rate in the high- and low-frequency regions has increased to -12 dB/octave or -40 dB/decade. At f_1 and f_2, therefore, the decibel drop is now -6 dB rather than the defined band frequency gain level of -3 dB. The -3-dB point has shifted to f_1' and f_2' as indicated, with a resulting drop in the bandwidth. A -18-dB/octave or -60-dB/decade slope will result for a three-stage system of identical stages with the indicated reduction in bandwidth (f_1'' and f_2'').

Assuming identical stages, an equation for each band frequency as a function of the number of stages (n) can be determined in the following manner: For the low-frequency region,

$$A_{v_{\text{low, (overall)}}} = A_{v_{1_{\text{low}}}} A_{v_{2_{\text{low}}}} A_{v_{3_{\text{low}}}} \cdots A_{v_{n_{\text{low}}}}$$

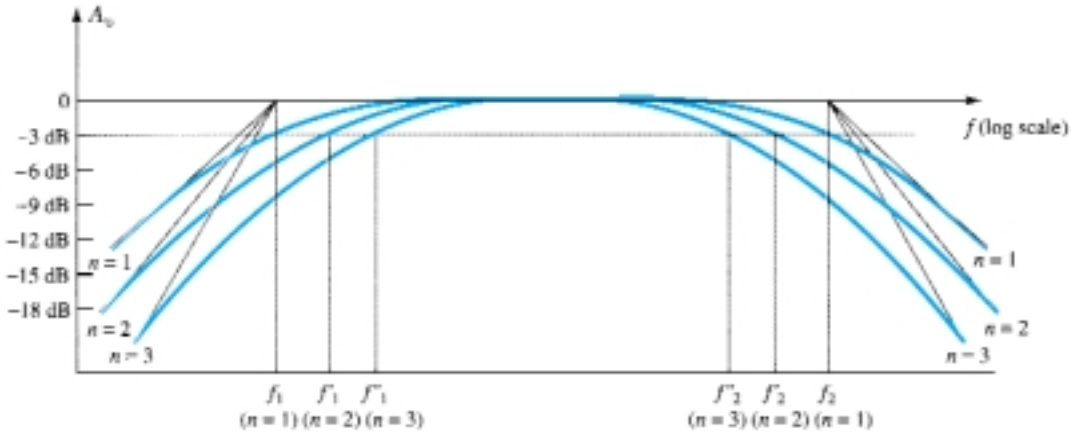


Figure 11.57 Effect of an increased number of stages on the cutoff frequencies and the bandwidth.

but since each stage is identical, $A_{v_{1_{low}}} = A_{v_{2_{low}}} = \text{etc.}$ and

$$A_{v_{low,\,(overall)}} = (A_{v_{1_{low}}})^n$$

or
$$\frac{A_{v_{low}}}{A_{v_{mid}}}\,(\text{overall}) = \left(\frac{A_{v_{1_{low}}}}{A_{v_{mid}}}\right)^n = \frac{1}{(1 - jf_1/f)^n}$$

Setting the magnitude of this result equal to $1/\sqrt{2}$(−3 dB level) results in

$$\frac{1}{\sqrt{1 + (f_1/f_1')^2]^n}} = \frac{1}{\sqrt{2}}$$

or
$$\left\{\left[1 + \left(\frac{f_1}{f_1'}\right)^{2-}\right]^{1/2}\right\}^n = \left\{\left[1 + \left(\frac{f_1}{f_1'}\right)^2\right]^n\right\}^{1/2} = (2)^{1/2}$$

so that
$$\left[1 + \left(\frac{f_1}{f_1'}\right)^2\right]^n = 2$$

and
$$1 + \left(\frac{f_1}{f_1'}\right)^2 = 2^{1/n}$$

with the result that
$$\boxed{f_1' = \frac{f_1}{\sqrt{2^{1/n} - 1}}} \tag{11.66}$$

In a similar manner, it can be shown that for the high-frequency region,

$$\boxed{f_2' = (\sqrt{2^{1/n} - 1})f_2} \tag{11.67}$$

Note the presence of the same factor $\sqrt{2^{1/n} - 1}$ in each equation. The magnitude of this factor for various values of n is listed below.

n	$\sqrt{2^{1/n} - 1}$
2	0.64
3	0.51
4	0.43
5	0.39

For $n = 2$, consider that the upper cutoff frequency $f_2' = 0.64f_2$ or 64% of the value obtained for a single stage, while $f_1' = (1/0.64)f_1 = 1.56f_1$. For $n = 3$, $f_2' = 0.51f_2$ or approximately $\frac{1}{2}$ the value of a single stage with $f_1' = (1/0.51)f_1 = 1.96f_1$ or approximately *twice* the single-stage value.

For the *RC*-coupled transistor amplifier, if $f_2 = f_\beta$, or if they are close enough in magnitude for both to affect the upper 3-dB frequency, the number of stages must be increased by a factor of 2 when determining f_2' due to the increased number of factors $1/(1 + jf/f_x)$.

A decrease in bandwidth is not always associated with an increase in the number of stages if the midband gain can remain fixed and independent of the number of stages. For instance, if a single-stage amplifier produces a gain of 100 with a bandwidth of 10,000 Hz, the resulting gain–bandwidth product is $10^2 \times 10^4 = 10^6$. For a two-stage system the same gain can be obtained by having two stages with a gain of 10 since ($10 \times 10 = 100$). The bandwidth of each stage would then increase by a factor of 10 to 100,000 due to the lower gain requirement and fixed gain–bandwidth product of 10^6. Of course, the design must be such as to permit the increased bandwidth and establish the lower gain level.

11.12 SQUARE-WAVE TESTING

A sense for the frequency response of an amplifier can be determined experimentally by applying a square-wave signal to the amplifier and noting the output response. The shape of the output waveform will reveal whether the high or low frequencies are being properly amplified. The use of *square-wave testing* is significantly less time-consuming than applying a series of sinusoidal signals at different frequencies and magnitudes to test the frequency response of the amplifier.

The reason for choosing a square-wave signal for the testing process is best described by examining the *Fourier series* expansion of a square wave composed of a series of sinusoidal components of different magnitudes and frequencies. The summation of the terms of the series will result in the original waveform. In other words, even though a waveform may not be sinusoidal, it can be reproduced by a series of sinusoidal terms of different frequencies and magnitudes.

The Fourier series expansion for the square wave of Fig. 11.58 is

$$v = \frac{4}{\pi} V_m \left(\sin 2\pi f_s t + \frac{1}{3} \sin 2\pi(3f_s)t + \frac{1}{5} \sin 2\pi(5f_s)t + \frac{1}{7} \sin 2\pi(7f_s)t + \frac{1}{9} \sin 2\pi(9f_s)t + \cdots + \frac{1}{n} \sin 2\pi(nf_s)t \right) \qquad (11.68)$$

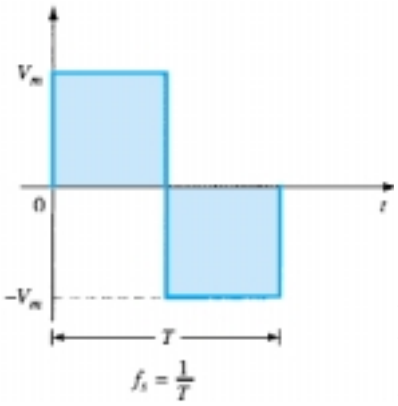


Figure 11.58 Square wave.

The first term of the series is called the *fundamental* term and in this case has the same frequency, f_s, as the square wave. The next term has a frequency equal to three times the fundamental and is referred to as the *third harmonic*. Its magnitude is one-third the magnitude of the fundamental term. The frequencies of the succeeding terms are odd multiples of the fundamental term, and the magnitude decreases with each higher harmonic. Figure 11.59 demonstrates how the summation of terms of a Fourier series can result in a nonsinusoidal waveform. The generation of the square wave of Fig. 11.58 would require an infinite number of terms. However, the summation of just the fundamental term and the third harmonic in Fig. 11.59a clearly results in a waveform that is beginning to take on the appearance of a square wave. Including the fifth and seventh harmonics as in Fig. 11.59b takes us a step closer to the waveform of Fig. 11.58.

Since the ninth harmonic has a magnitude greater than 10% of the fundamental term [$\frac{1}{9}(100\%) = 11.1\%$], the fundamental term through the ninth harmonic are the major contributors to the Fourier series expansion of the square-wave function. It is therefore reasonable to assume that if the application of a square wave of a particular frequency results in a nice clean square wave at the output, then the fundamental

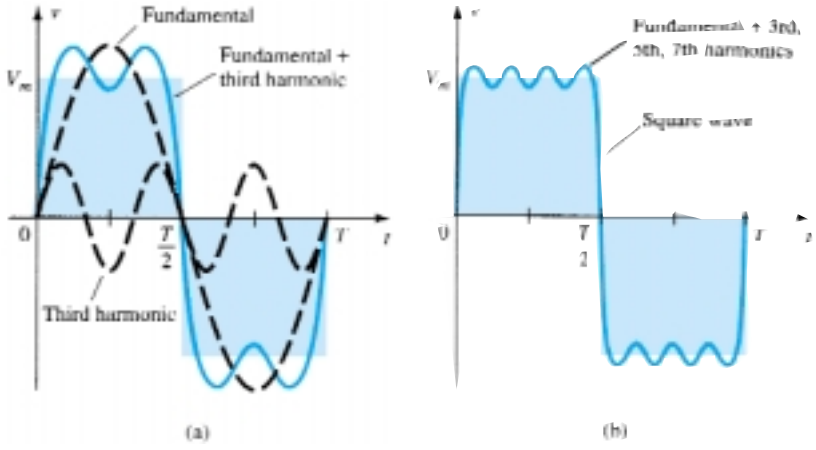


Figure 11.59 Harmonic content of a square wave.

through the ninth harmonic are being amplified without visual distortion by the amplifier. For instance, if an audio amplifier with a bandwidth of 20 kHz (audio range is from 20 Hz to 20 kHz) is to be tested, the frequency of the applied signal should be at least 20 kHz/9 = 2.22 kHz.

If the response of an amplifier to an applied square wave is an undistorted replica of the input, the frequency response (or BW) of the amplifier is obviously sufficient for the applied frequency. If the response is as shown in Fig. 11.60a and b, the low frequencies are not being amplified properly and the low cutoff frequency has to be investigated. If the waveform has the appearance of Fig. 11.60c, the high-frequency components are not receiving sufficient amplification and the high cutoff frequency (or BW) has to be reviewed.

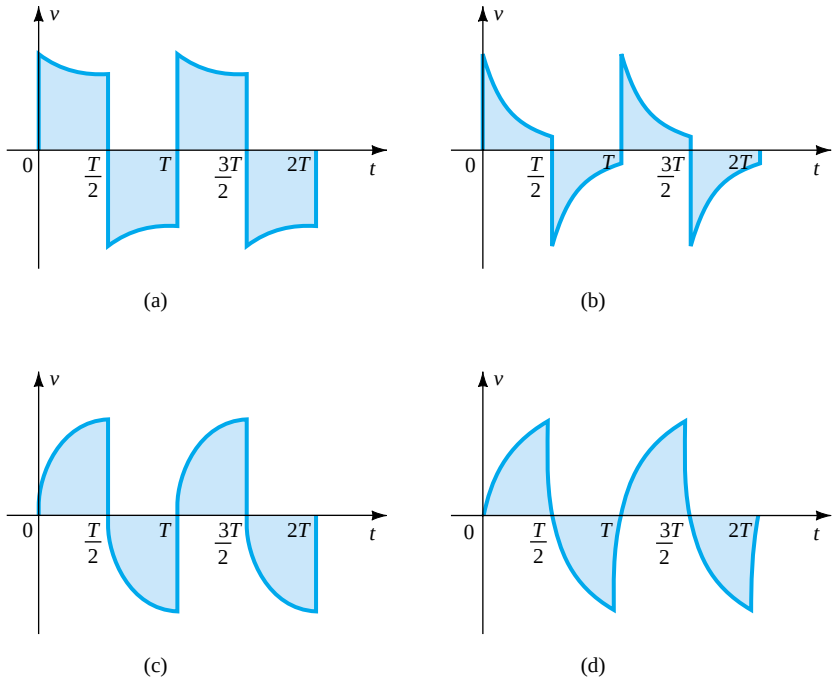


Figure 11.60 (a) Poor low frequency response; (b) very poor low-frequency response; (c) poor high-frequency response; (d) very poor high-frequency response.

The actual high cutoff frequency (or BW) can be determined from the output waveform by carefully measuring the rise time defined between 10% and 90% of the peak value, as shown in Fig. 11.61. Substituting into the following equation will provide the upper cutoff frequency, and since BW = $f_{H_i} - f_{L_o} \cong f_{H_i}$, the equation also provides an indication of the BW of the amplifier.

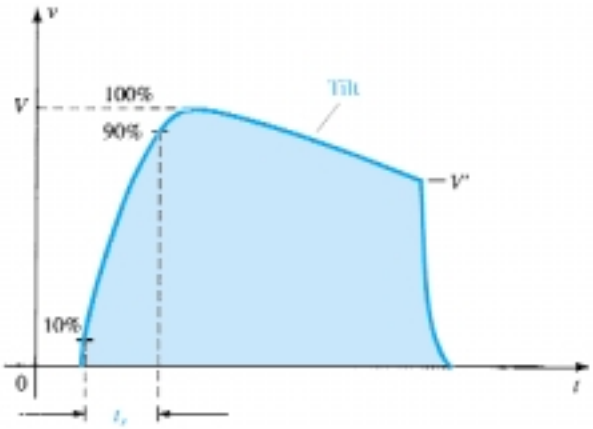


Figure 11.61 Defining the rise time and tilt of a square wave response.

f

$$\text{BW} \cong f_{H_i} = \frac{0.35}{t_r} \tag{11.69}$$

The low cutoff frequency can be determined from the output response by carefully measuring the tilt of Fig. 11.61 and substituting into one of the following equations:

$$\%\ \text{tilt} = P\% = \frac{V - V'}{V} \times 100\% \tag{11.70}$$

$$\text{tilt} = P = \frac{V - V'}{V} \quad \text{(decimal form)} \tag{11.71}$$

The low cutoff frequency is then determined from

$$f_{L_o} = \frac{P}{\pi} f_s \tag{11.72}$$

EXAMPLE 11.13

The application of a 1-mV, 5-kHz square wave to an amplifier resulted in the output waveform of Fig. 11.62.

(a) Write the Fourier series expansion for the square wave through the ninth harmonic.

(b) Determine the bandwidth of the amplifier.

(c) Calculate the low cutoff frequency.

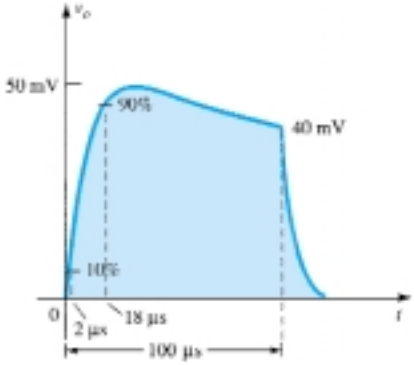


Figure 11.62 Example 11.13

Solution

(a) $$v_i = \frac{4\ \text{mV}}{\pi}\left(\sin 2\pi\,(5 \times 10^3)t + \frac{1}{3}\sin 2\pi(15 \times 10^3)t + \frac{1}{5}\sin 2\pi(25 \times 10^3)t + \frac{1}{7}\sin 2\pi(35 \times 10^3)t + \frac{1}{9}\sin 2\pi(45 \times 10^3)t\right)$$

(b) $$t_r = 18\ \mu\text{s} - 2\ \mu\text{s} = 16\ \mu\text{s}$$

$$\text{BW} = \frac{0.35}{t_r} = \frac{0.35}{16\ \mu\text{s}} = \mathbf{21{,}875\ Hz} \cong 4.4 f_s$$

(c) $$P = \frac{V - V'}{V} = \frac{50\ \text{mV} - 40\ \text{mV}}{50\ \text{mV}} = 0.2$$

$$f_{L_o} = \frac{P}{\pi} f_s = \left(\frac{0.2}{\pi}\right)(5\ \text{kHz}) = \mathbf{318.31\ Hz}$$

11.13 PSPICE WINDOWS

The computer analysis of this chapter was integrated into the chapter for emphasis and a clear demonstration of the power of the PSpice software package. The complete frequency response of a single-stage or multistage system can be determined in a relatively short period of time to verify theoretical calculations or provide an immediate indication of the low and high cutoff frequencies of the system. The exercises in the chapter will provide an opportunity to apply the PSpice software package to a variety of networks.

§ 11.2 Logarithms

1. (a) Determine the common logarithm of the following numbers: 10^3, 50, and 0.707.
(b) Determine the natural logarithm of the same numbers appearing in part (a).
(c) Compare the solutions of parts (a) and (b).

2. (a) Determine the common logarithm of the number 2.2×10^3.
(b) Determine the natural logarithm of the number of part (a) using Eq. (11.4).
(c) Determine the natural logarithm of the number of part (a) using natural logarithms and compare with the solution of part (b).

3. Determine:
(a) $20 \log_{10} \frac{40}{8}$ using Eq. (11.6) and compare with $20 \log_{10} 5$.
(b) $10 \log_{10} \frac{1}{20}$ using Eq. (11.7) and compare with $10 \log_{10} 0.05$.
(c) $\log_{10}(40)(0.125)$ using Eq. (11.8) and compare with $\log_{10} 5$.

4. Calculate the power gain in decibels for each of the following cases.
(a) $P_o = 100$ W, $P_i = 5$ W.
(b) $P_o = 100$ mW, $P_i = 5$ mW.
(c) $P_o = 100\ \mu$W, $P_i = 20\ \mu$W.

5. Determine G_{dBm} for an output power level of 25 W.

6. Two voltage measurements made across the same resistance are $V_1 = 25$ V and $V_2 = 100$ V. Calculate the power gain in decibels of the second reading over the first reading.

7. Input and output voltage measurements of $V_i = 10$ mV and $V_o = 25$ V are made. What is the voltage gain in decibels?

* **8.** (a) The total decibel gain of a three-stage system is 120 dB. Determine the decibel gain of each stage if the second stage has twice the decibel gain of the first and the third has 2.7 times the decibel gain of the first.
(b) Determine the voltage gain of each stage.

* **9.** If the applied ac power to a system is 5 μW at 100 mV and the output power is 48 W, determine:
(a) The power gain in decibels.
(b) The voltage gain in decibels if the output impedance is 40 kΩ.
(c) The input impedance.
(d) The output voltage.

§ 11.4 General Frequency Considerations

10. Given the characteristics of Fig. 11.63, sketch:
(a) The normalized gain.
(b) The normalized dB gain (and determine the bandwidth and cutoff frequencies).

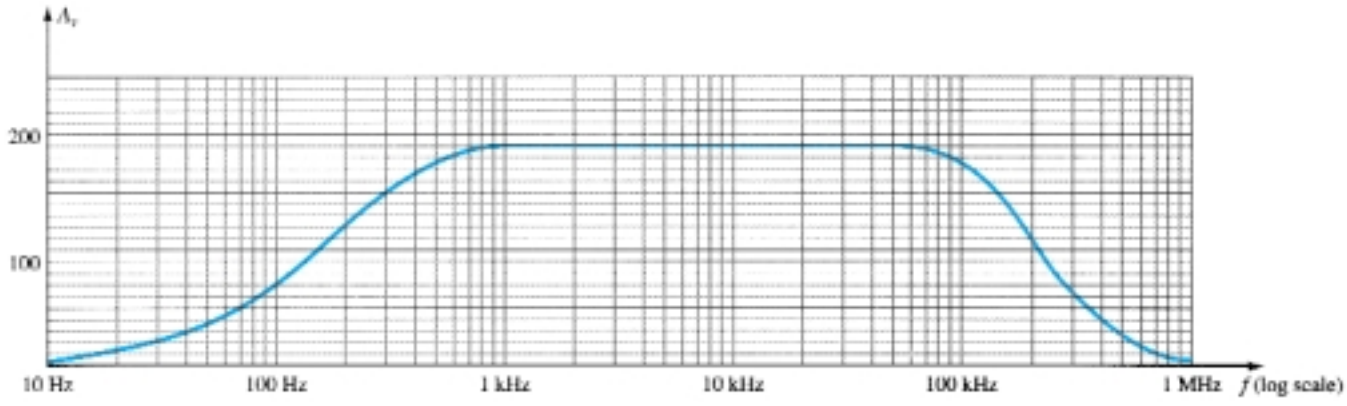


Figure 11.63 Problem 10

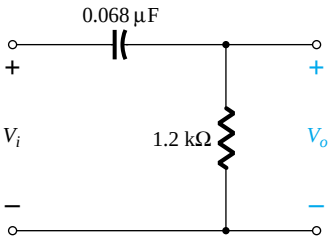


Figure 11.64 Problems 11, 12, and 32

§ 11.5 Low-Frequency Analysis—Bode Plot

11. For the network of Fig. 11.64:
(a) Determine the mathematical expression for the magnitude of the ratio V_o/V_i.
(b) Using the results of part (a), determine V_o/V_i at 100 Hz, 1 kHz, 2 kHz, 5 kHz, and 10 kHz, and plot the resulting curve for the frequency range of 100 Hz to 10 kHz. Use a log scale.
(c) Determine the break frequency.
(d) Sketch the asymptotes and locate the −3-dB point.
(e) Sketch the frequency response for V_o/V_i and compare to the results of part (b).

12. For the network of Fig. 11.64:
(a) Determine the mathematical expression for the angle by which V_o leads V_i.
(b) Determine the phase angle at f = 100 Hz, 1 kHz, 2 kHz, 5 kHz, and 10 kHz, and plot the resulting curve for the frequency range of 100 Hz to 10 kHz.
(c) Determine the break frequency.
(d) Sketch the frequency response of θ for the same frequency spectrum of part (b) and compare results.

13. (a) What frequency is 1 octave above 5 kHz?
(b) What frequency is 1 decade below 10 kHz?
(c) What frequency is 2 octaves below 20 kHz?
(d) What frequency is 2 decades above 1 kHz?

§ 11.6 Low-Frequency Response — BJT Amplifier

14. Repeat the analysis of Example 11.9 with r_o = 40 kΩ. What is the effect on $A_{v_{mid}}$, f_{L_S}, f_{L_C}, f_{L_E}, and the resulting cutoff frequency?

15. For the network of Fig. 11.65:
(a) Determine r_e.
(b) Find $A_{v_{mid}} = V_o/V_i$.
(c) Calculate Z_i.
(d) Find $A_{v_{Smid}} = V_o/V_s$.
(e) Determine f_{L_S}, f_{L_C}, and f_{L_E}.
(f) Determine the low cutoff frequency.
(g) Sketch the asymptotes of the Bode plot defined by the cutoff frequencies of part (e).
(h) Sketch the low-frequency response for the amplifier using the results of part (f).

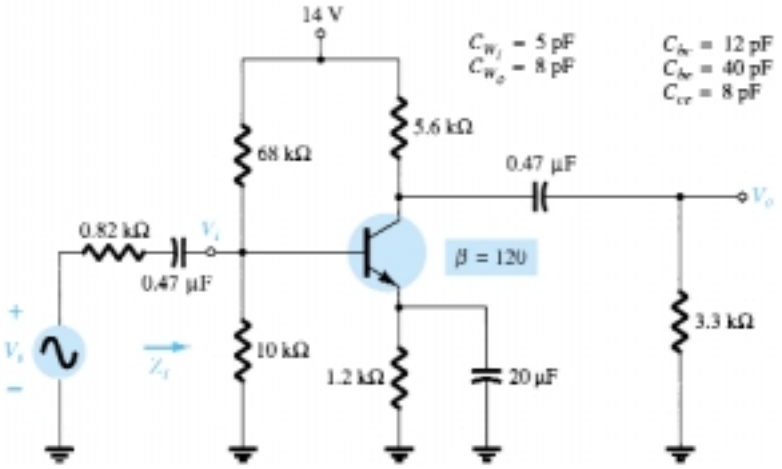


Figure 11.65 Problems 15, 22, and 33

* **16.** Repeat Problem 15 for the emitter-stabilized network of Fig. 11.66.

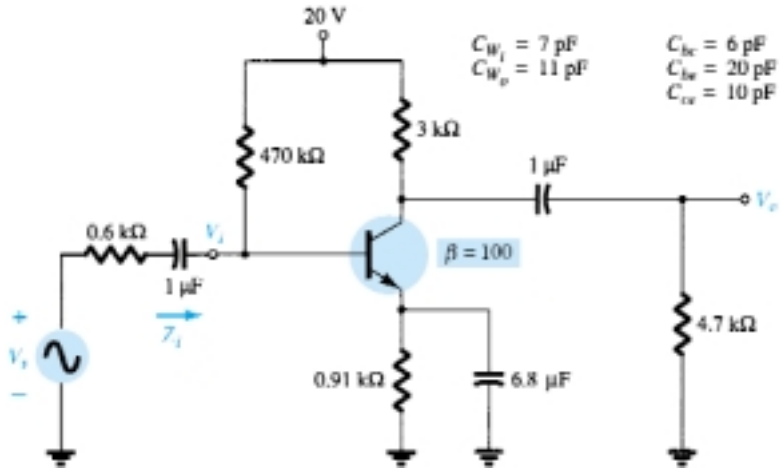


Figure 11.66 Problems 16 and 23

* **17.** Repeat Problem 15 for the emitter-follower network of Fig. 11.67.

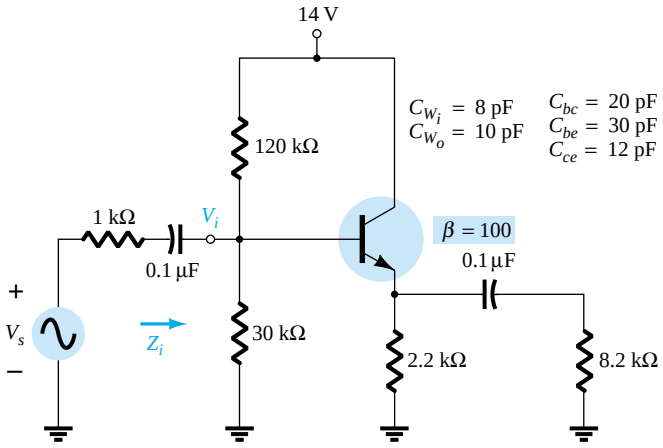


Figure 11.67 Problems 17 and 24

* **18.** Repeat Problem 15 for the common-base configuration of Fig. 11.68. Keep in mind that the common-base configuration is a noninverting network when you consider the Miller effect.

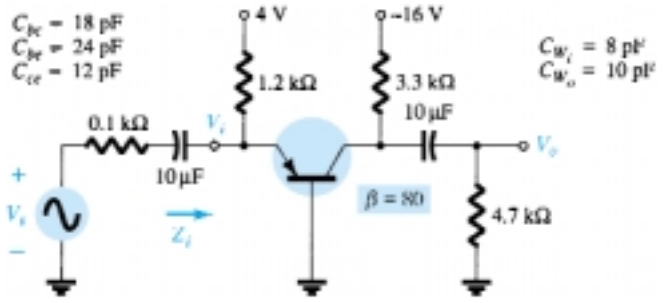


Figure 11.68 Problems 18, 25, and 34

§ 11.7 Low-Frequency Response — FET Amplifier

19. For the network of Fig. 11.69:
(a) Determine V_{GS_Q} and I_{D_Q}.
(b) Find g_{m0} and g_m.
(c) Calculate the midband gain of $A_v = V_o/V_i$.
(d) Determine Z_i.
(e) Calculate $A_{v_s} = V_o/V_s$.
(f) Determine f_{L_G}, f_{L_C}, and f_{L_S}.
(g) Determine the low cutoff frequency.
(h) Sketch the asymptotes of the Bode plot defined by part (f).
(i) Sketch the low-frequency response for the amplifier using the results of part (f).

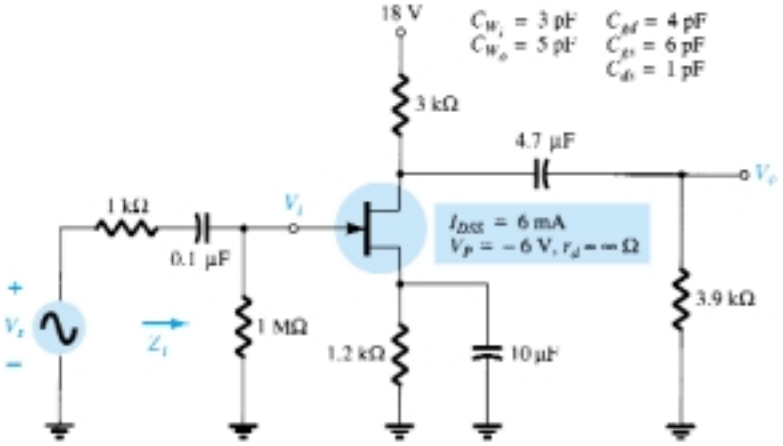


Figure 11.69 Problems 19, 20, 26, and 35

* **20.** Repeat the analysis of Problem 19 with $r_d = 100$ kΩ. Does it have an impact of any consequence on the results? If so, which elements?

* **21.** Repeat the analysis of Problem 19 for the network of Fig. 11.70. What effect did the voltage-divider configuration have on the input impedance and the gain A_{v_s} compared to the biasing arrangement of Fig. 11.69?

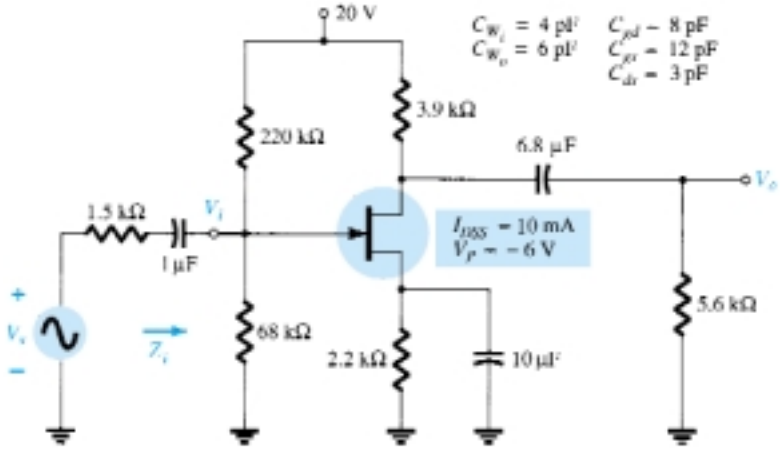


Figure 11.70 Problems 21 and 27

§ 11.9 High-Frequency Response — BJT Amplifier

22. For the network of Fig. 11.65:
(a) Determine f_{H_i} and f_{H_o}.
(b) Assuming that $C_{b'e} = C_{be}$ and $C_{b'c} = C_{bc}$, find f_β and f_T.
(c) Sketch the frequency response for the high-frequency region using a Bode plot and determine the cutoff frequency.

* **23.** Repeat the analysis of Problem 22 for the network of Fig. 11.66.

* **24.** Repeat the analysis of Problem 22 for the network of Fig. 11.67.

* **25.** Repeat the analysis of Problem 22 for the network of Fig. 11.68.

§ 11.10 High-Frequency Response — FET Amplifier

26. For the network of Fig. 11.69:
(a) Determine g_{m_0} and g_m.
(b) Find A_v and A_{v_s} in the mid-frequency range.
(c) Determine f_{H_i} and f_{H_o}.
(d) Sketch the frequency response for the high-frequency region using a Bode plot and determine the cutoff frequency.

* **27.** Repeat the analysis of Problem 26 for the network of Fig. 11.70.

§ 11.11 Multistage Frequency Effects

28. Calculate the overall voltage gain of four identical stages of an amplifier, each having a gain of 20.

29. Calculate the overall upper 3-dB frequency for a four-stage amplifier having an individual stage value of $f_2 = 2.5$ MHz.

30. A four-stage amplifier has a lower 3-dB frequency for an individual stage of $f_1 = 40$ Hz. What is the value of f_1 for this full amplifier?

§ 11.12 Square-Wave Testing

* **31.** The application of a 10-mV, 100-kHz square wave to an amplifier resulted in the output waveform of Fig. 11.71.
(a) Write the Fourier series expansion for the square wave through the ninth harmonic.
(b) Determine the bandwidth of the amplifier to the accuracy available by the waveform of Fig. 11.71.
(c) Calculate the low cutoff frequency.

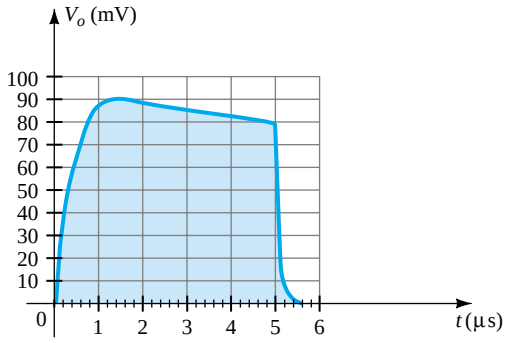


Figure 11.71 Problem 31

§ 11.13 PSpice Windows

32. Using PSpice Windows, determine the frequency response of V_o/V_i for the high-pass filter of Fig. 11.64.

33. Using PSpice Windows, determine the frequency response of V_o/V_s for the BJT amplifier of Fig. 11.65.

34. Repeat Problem 33 for the network of Fig. 11.68.

35. Repeat Problem 33 for the JFET configuration of Fig. 11.69.

*Please Note: Asterisks indicate more difficult problems.